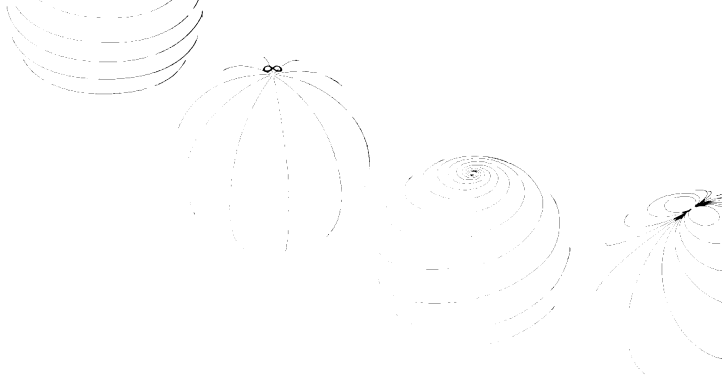




CHAPTER 7

Winding Numbers and Topology

In this chapter we shall investigate a simple but immensely powerful concept—the number of times a loop winds around a point. In Chapter 2 we saw that this concept was needed to understand multifunctions, and in the next chapter we will see that it plays an equally crucial role in understanding complex integration. However, only the first two sections [up to (7.2)] of the present chapter are actually a prerequisite for that work; the rest may be read at any time. If you are in a rush to learn about integration, you may wish to skip the rest of the chapter and return to it later.

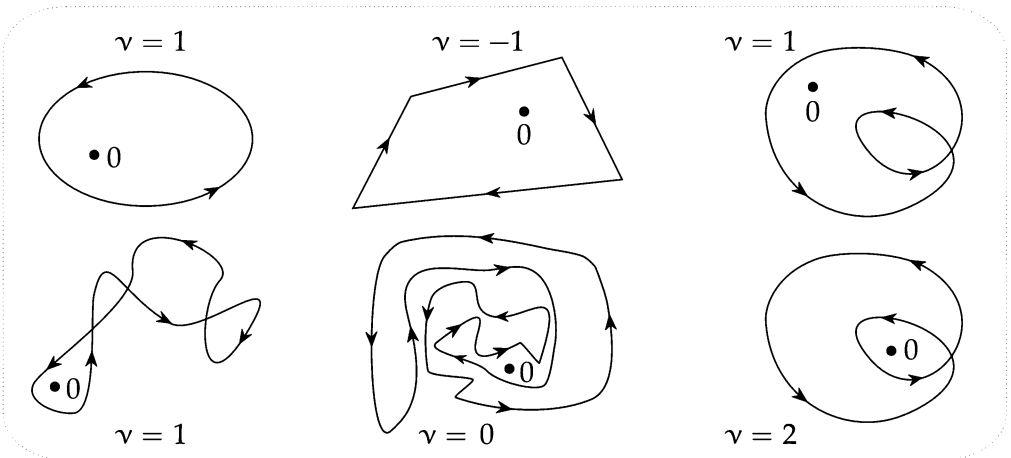
7.1 Winding Number

7.1.1 The Definition

As the name suggests, the *winding number* $v(L, 0)$ of a closed loop L about the origin 0 is simply the *net* number of revolutions of the direction of z as it traces out L once in its given sense. A nut on a bolt admirably illustrates the concept of “net rotation”: spin the nut this way and that way for a while; the final distance of the nut from its starting point measures the net rotation it has undergone.

Figure [7.1] shows six loops and their corresponding winding numbers. You can verify these values by starting at a random point on each curve and tracing it out with your finger: starting with zero, add one after each positive (= counterclockwise) revolution of the vector connecting the origin to your finger, and subtract one after each negative (= clockwise) revolution. When you have returned to your starting point, the final count is the winding number of the loop.

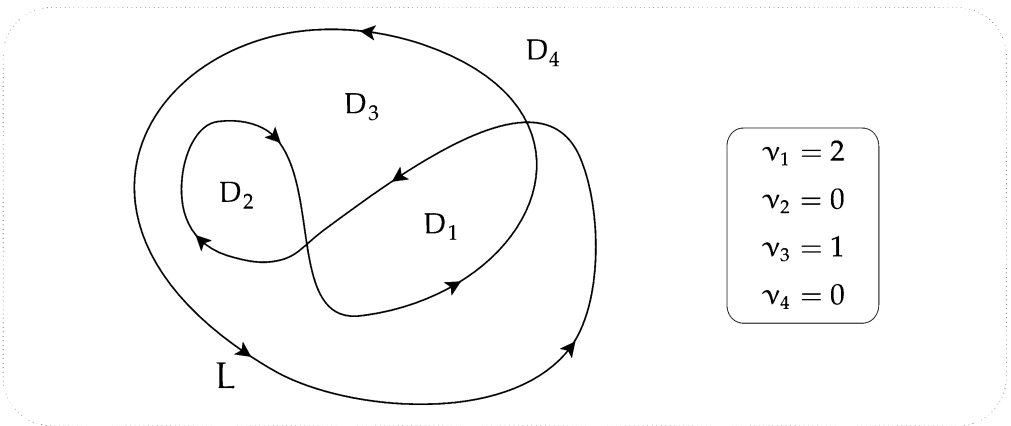
It is often useful to consider the winding number of a loop about a point p other than the origin, and this is correspondingly written $v(L, p)$. Instead of counting the revolutions of z , we now count those of $(z - p)$. For example, the shaded region in [7.1] can be defined as all the positions of p for which $v(L, p) \neq 0$. Try shading this set for the other loops.



[7.1] The **winding number** ν of a closed loop L counts the number of times L encircles a point.

7.1.2 What Does “Inside” Mean?

A loop is called *simple* if it does not intersect itself; for example, circles, ellipses, and triangles are all simple. Although a simple loop can actually be very complicated [see Ex. 1] it seems clear, though it is hard to prove, that it will divide the plane into just two sets, its inside and its outside. However, in the case of a loop that is not simple, such as [7.2], it is no longer obvious which points are to be considered inside the loop, and which outside. The winding number concept allows us make the desired distinction clearly.



[7.2] Using the winding number to define the “inside” and the “outside” of a closed loop. A non-simple loop L partitions the plane into multiple simply connected regions, D_j , and the winding number is constant in each one, taking the value ν_j . The “inside” of L is then defined to be the union $D_1 \cup D_3$ of the D_j ’s with $\nu_j \neq 0$, and the “outside” is the union $D_2 \cup D_4$ of the D_j ’s with $\nu_j = 0$.

A typical loop such as L will partition the plane into a number of sets D_j (four in this case). If the point p wanders around within one of these sets then it seems plausible that the winding number $v(L, p)$ remains constant. Let's check this.

Concentrate on just a short segment of L . As z traverses it, the rotation of $(z - p)$ will depend continuously on p unless¹ p crosses L . In other words, if we move p a tiny bit then the rotation angle will likewise only change a tiny bit. Since the winding number of L is just the sum of the rotations due to all its segments, it follows that it too depends continuously on the location of p : a tiny movement of p to $\tilde{p}$ can only produce a tiny change $[v(L, \tilde{p}) - v(L, p)]$ in the winding number. But since this small difference is an *integer*, it must be exactly 0. Done.

Since L winds round each point of D_j the same number of times, it follows that we can attach a winding number v_j to the set as a whole. Verify the values of v_j given in the figure.

The “inside” can now be *defined* to consist of those D_j for which $v_j \neq 0$, while the remaining D_j constitute the “outside”. Thus in [7.2] we find that $D_1 \cup D_3$ is the inside, while $D_2 \cup D_4$ is the outside.

The “correctness” of this definition will become apparent in the next chapter.

7.1.3 Finding Winding Numbers Quickly

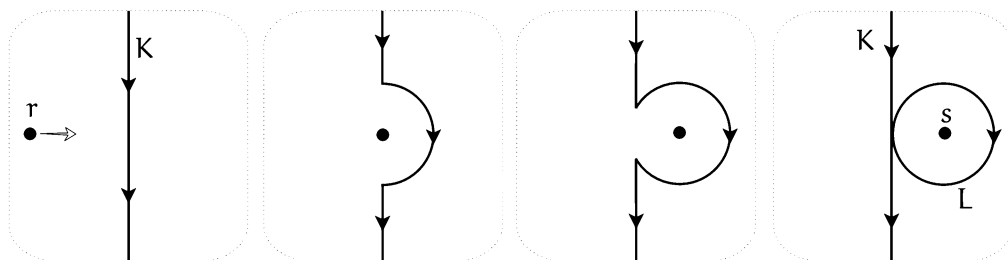
In [7.2] we found the winding numbers directly from the definition: we strenuously followed the curve with our finger (or eye) and counted revolutions. For a really complicated loop this could literally become a headache. We now derive a much quicker and more elegant method of visually computing winding numbers.

If a point r moves around without crossing a loop K then $v(K, r)$ remains constant, but what happens when the point *does* cross K ? Consider [7.3]. On the far left, close to the loop K , is the point r ; the rest of K is off the picture, and the number of times it winds round r is $v(K, r)$. The time-lapse pictures in [7.3] show r moving towards the loop, which itself deforms so as to avoid being crossed, finally ending up at the point s .

Now since the moving point never crosses the loop, the winding number remains constant throughout the process. But on the far right, the new loop can be thought of as the union of the old loop K , together with the new circle L . Thus,

$$\begin{aligned} v(K, r) &= v(K, s) + v(L, s) = v(K, s) - 1 \\ \Rightarrow v(K, s) &= v(K, r) + 1. \end{aligned}$$

¹ Consider the behaviour of the rotation due to a short segment of L as p crosses it.



[7.3] Crossing Rule: If the loop is travelling from our left to our right as we cross it, $v \rightsquigarrow v + 1$. For imagine that K deforms and moves out of our way, so that we never cross it, then the winding number does not change. But on the far right, K has evolved into K and a clockwise circle, L , so $v(K, r) = v(K, s) + v(L, s) = v(K, s) - 1$, and therefore $v(K, s) = v(K, r) + 1$, as claimed.

Imagining ourselves at r , looking towards K as we approach it, we may express this result in the form of the following very useful *crossing rule*:

If K is moving from our left to our right [our right to our left] as we cross it, its winding number around us increases [decreases] by one. (7.1)

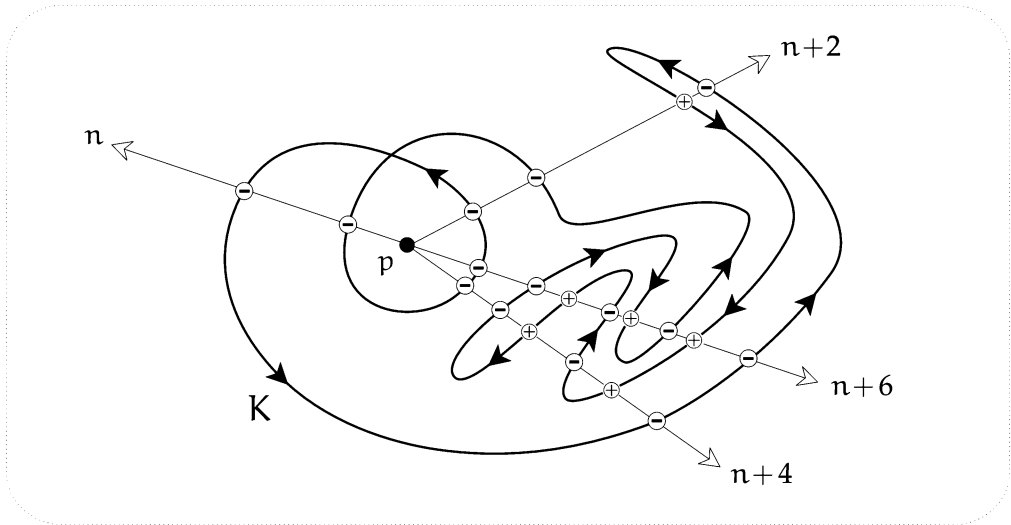
Using this result, it is incredibly quick and easy to find the v_j 's for even the most complicated loop. Try it out on [7.2]. Starting your journey well outside L , where you know that the winding number is zero, move from region to region, using crossing rule (7.1) to add or subtract one at each crossing of L .

An immediate consequence of this idea is a connection between $n = v(K, p)$ and the number of intersection points of K with a ray emanating from p . Suppose that the ray is in general position in the sense that it doesn't pass through any self-intersection points of K , nor is it tangent to K . If a point q on this ray is sufficiently distant from p then clearly K cannot wind around it; thus as we move along the ray from p to q the winding number changes by n . But the winding number only changes when we cross K , and only one unit per crossing. The ray must therefore intersect K at least $|n|$ times. However, in addition to these $|n|$ necessary crossings there may be additional cancelling *pairs* of crossings. In general, then, the number of intersection points will be $|n|$, or $|n| + 2$, or $|n| + 4$, etc. Figure [7.4] illustrates these possibilities for a case in which $n = 2$, each intersection point being marked with $\oplus$ or $\ominus$ according as the winding number increases or decreases as it is crossed.

7.2 Hopf's Degree Theorem

7.2.1 The Result

We have discussed the fact that for a fixed loop and a continuously moving point, the winding number only changes when the point crosses the loop. But it is clear



[7.4] A ray emanating from p intersects K at least $|\nu(K, p)|$ times. For as we move along the ray towards infinity, the winding number must eventually drop to zero. But the initial value $n = \nu(K, p)$ only changes when we cross K , and only one unit at a time, so there must be at least $|n|$ such crossings. But, in addition to these $|n|$ necessary crossings, there may be additional cancelling pairs of crossings.

that the same must be true of a fixed point and a continuously moving loop: the winding number of the evolving loop can only change if it crosses the point, and it changes by ± 1 according to the same crossing rule as before. Thus if a loop K can be continuously deformed into another loop L without ever crossing a point p , the winding numbers of K and L around p will be equal.

It is natural to ask if the converse is also true: if K and L wind round p the same number of times, is it always possible to deform K into L without ever crossing p ? This is certainly a more subtle question, but by drawing examples you will be led to suspect that it is true. In this section we will confirm this hunch, so establishing that

A loop K may be continuously deformed into another loop L , without ever crossing the point p , if and only if K and L have the same winding number round p . (7.2)

At the end of the next chapter, this will turn out to be the key to understanding one of the central results of complex analysis.

The result in (7.2) is the simplest example of a remarkable topological fact, called *Hopf's Degree Theorem*, that is valid in any number of dimensions. In the 2-dimensional complex plane, a point can be surrounded using a closed 1-dimensional curve—a loop. In 3-dimensional space, a point can be surrounded using a closed 2-dimensional surface. Just as a circle in the plane winds once

around its centre, so a sphere in space encloses its centre just once. More generally, self-intersecting loops in the plane may enclose a point several times, and this is precisely what ν counts. Similarly, it is possible to define a more general concept (*degree*) that counts the number of times a surface surrounds a point in space. Hopf's Theorem now says that one closed surface may be continuously deformed into another, without ever crossing p , if and only if they enclose p the same number of times. Indeed, Hopf's Theorem says the same is true of n -dimensional surfaces enclosing points in $(n + 1)$ -dimensional space!

7.2.2 Loops as Mappings of the Circle*

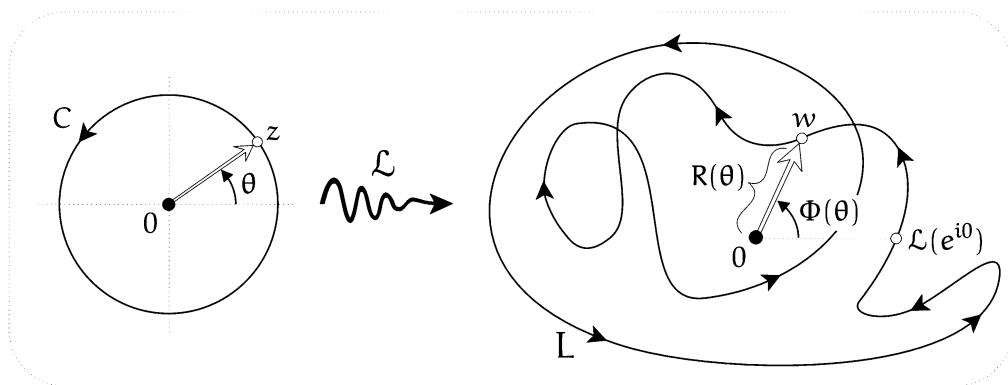
As a first step to understanding (7.2), we will look at loops in a new way. Let C be a rubber band in the shape of the unit circle. We may now deform C into the shape of any desired loop L . At the end of the deformation process, each point z of C has been brought to a definite image point w on L , and thus L may be thought of as the image of C under a continuous mapping $w = \mathcal{L}(z)$. See [7.5].

As θ varies from 0 to 2π , $z = e^{i\theta}$ moves round C once and w moves round L once, the length R and angle Φ of w varying continuously with θ . We may write

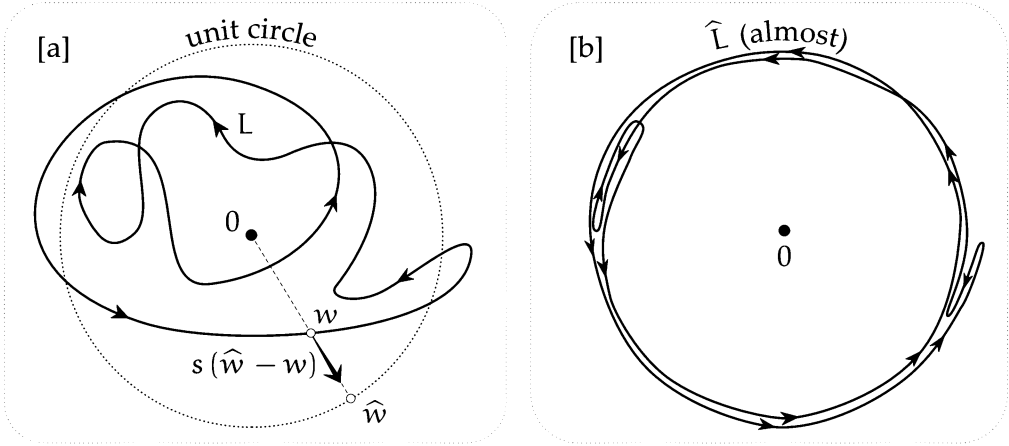
$$w = \mathcal{L}(e^{i\theta}) = R(\theta) e^{i\Phi(\theta)},$$

where $R(\theta)$ and $\Phi(\theta)$ are continuous functions. By rotating L (if necessary) we can ensure that $\mathcal{L}(e^{i0})$ is a positive real number, so that we may set $\Phi(0) = 0$. The net rotation of w after it has returned to its starting point is then given by $\Phi(2\pi) = 2\pi\nu$.

Clearly, the varying length of w is something of a red herring when it comes to understanding winding numbers, and we now remove this distraction by pulling each point w of L radially onto the point $\hat{w} = w/|w|$ on the unit circle, so obtaining



[7.5] A closed loop L may be viewed as the image of the unit circle C under a continuous mapping $\mathcal{L}$. As θ goes from 0 to 2π , $z = e^{i\theta}$ traverses C once, and $w = \mathcal{L}(e^{i\theta}) = R(\theta) e^{i\Phi(\theta)}$ traverses L once. If $\Phi(0) = 0$ then the net rotation of w after it has returned to its starting point is given by $\Phi(2\pi) = 2\pi\nu$.



[7.6] [a] The loop L can be gradually pushed onto the unit circle by moving w along $(\hat{w} - w)$. As s increases from 0 to 1, $\mathcal{L}_s(z) = w + s(\hat{w} - w)$ moves radially from w to $\hat{w}$ on the unit circle. **[b]** With a value of s close to 1, the loop has almost evolved into its final form, $\hat{L}$, given by $\hat{w} = \hat{L}(e^{i\theta}) = e^{i\Phi(\theta)}$.

a standardized representation $\hat{L}$. We can even give an explicit prescription for gradually deforming L into $\hat{L}$. See [7.6a]. Since $(\hat{w} - w)$ is the complex number from a point on L to its destination on $\hat{L}$, the point that is a fraction s of the way there is

$$\mathcal{L}_s(z) = w + s(\hat{w} - w). \quad (7.3)$$

As s varies from 0 to 1, $\mathcal{L}_s(C)$ gradually (and reversibly) changes from L into $\hat{L}$. Figure [7.6b] shows $\mathcal{L}_s(C)$ for a value of s close to 1. Finally, note the obvious fact that as we gradually pull L radially onto $\hat{L}$, the origin is not crossed.

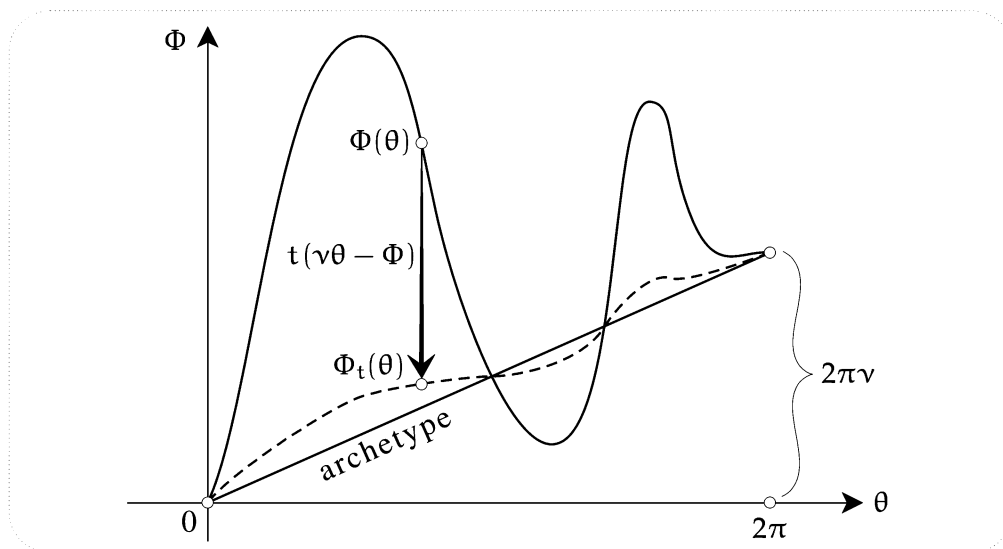
With lengths disposed of in this way, we are now dealing with a mapping $\hat{\mathcal{L}}$ from the unit circle C to the standardized loop $\hat{L}$ on the unit circle, where

$$\hat{w} = \hat{\mathcal{L}}(e^{i\theta}) = e^{i\Phi(\theta)}. \quad (7.4)$$

In this context, it is common to speak of the *degree* of the mapping $\hat{\mathcal{L}}$ which produces $\hat{L}$, rather than of the “winding number” of $\hat{L}$ (or L). The single real function $\Phi(\theta)$ completely describes the mapping $\hat{\mathcal{L}}$, and [7.7] shows how we can immediately read off the degree of $\hat{\mathcal{L}}$ (i.e. ν) from the graph of $\Phi(\theta)$. Make sure you are comfortable with the meaning of such a graph. For example, if z moves at unit speed round C , what does the slope (including the sign) of the graph represent?

7.2.3 The Explanation*

The archetypal mapping of degree ν is $\hat{\mathcal{J}}_\nu(z) = z^\nu$, for which $\Phi(\theta) = \nu\theta$. Its straight-line graph is shown in [7.7]. As z travels once round C at unit speed, $\hat{w}$ travels once round $\hat{\mathcal{J}}_\nu$ with speed $|\nu|$, completing $|\nu|$ circuits of the unit circle [counterclockwise if $\nu > 0$; clockwise if $\nu < 0$].



[7.7] Gradual evolution of a loop into the archetypal loop of the same winding number. If we imagine $\widehat{L}$ in the previous figure to be an elastic string wrapped around a cylinder, the slack will automatically be taken up, and the actual mapping $\widehat{L}$ of degree ν will evolve into the archetypal mapping of degree ν , namely, $\widehat{g}_\nu(z) = z^\nu$, for which $\Phi(\theta) = \nu\theta$. This can be achieved explicitly by following the illustrated evolution of $\Phi_t(\theta) = \Phi(\theta) + t[\nu\theta - \Phi(\theta)]$ from $t = 0$ to $t = 1$.

To see how a typical standardized loop $\widehat{L}$ of winding number ν is related to the archetypal loop $\widehat{J}_\nu$, we return to the example in [7.6b], for which $\nu = 2$. Thinking of the unit circle as the boundary of a solid cylinder, and recalling that the loop is an elastic band that wishes to contract, what will happen if we release $\widehat{L}$? The slack will be taken up, and $\widehat{L}$ will automatically contract itself into the archetypal loop $\widehat{J}_2$. This convincing mental image of the rubber band contracting into $\widehat{J}_\nu$ can be formalized to show that

Any $\widehat{L}$ of winding number ν can be continuously deformed into the archetypal loop $\widehat{J}_\nu$, and vice versa.

The process of “taking up the slack” can be explicitly described in terms of the graph of the Φ that describes $\widehat{L}$. As t varies from 0 to 1, the graph of

$$\Phi_t(\theta) = \Phi(\theta) + t[\nu\theta - \Phi(\theta)]$$

continuously and reversibly evolves from the graph of the general Φ into the straight-line graph of the archetype. The dashed curve in [7.7] is the graph of Φ_t for a value of t close to one. Defining

$$\widehat{\mathcal{L}}_t(e^{i\theta}) = e^{i\Phi_t(\theta)}, \quad (7.5)$$

$\widehat{\mathcal{L}}_t(C)$ therefore evolves continuously and reversibly from $\widehat{L}$ into $\widehat{J}_\nu$, as t varies from 0 to 1.

The explicit two-stage deformation given above [(7.3) followed by (7.5)] allows us to deform any loop of winding number ν into the archetypal loop $\widehat{J}_\nu$, and without the origin ever being crossed. Conversely, by reversing these steps, $\widehat{J}_\nu$ may be deformed into any loop of winding number ν . This demonstrates (7.2), for if K and L both have winding number ν , we may first deform K into $\widehat{J}_\nu$, and then deform $\widehat{J}_\nu$ into L .

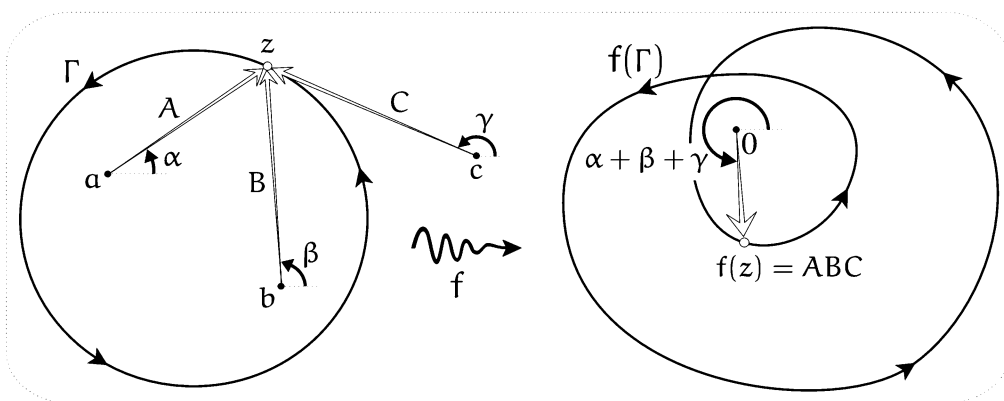
7.3 Polynomials and the Argument Principle

Let A , B , and C be the complex numbers from the fixed points a , b , and c to the variable point z . Figure [7.8] shows a circle Γ and its image $f(\Gamma)$ under the cubic mapping

$$f(z) = (z - a)(z - b)(z - c) = ABC.$$

Notice that Γ encircles two zeros of the mapping, while $f(\Gamma)$ has a winding number of 2 about zero. This is no accident. Since angles add when we multiply complex numbers, the number of revolutions executed by ABC is just the sum of the revolutions executed separately by each of A , B , and C . But as z goes round Γ once, A and B both execute a complete revolution, while the direction of C merely oscillates. Thus $\nu[f(\Gamma), 0] = 2$.

If we enlarged Γ so that it encircled c , then C would also execute a complete revolution, and the winding number would increase to 3. Once again, the number of points inside Γ that are mapped to 0 is the winding number of the image about that point.



[7.8] If a simple loop Γ winds once around m roots of a polynomial $P(z)$, then $\nu[P(\Gamma), 0] = m$. In the illustrated cubic example, the number of revolutions executed by $f(z) = ABC$ is just the sum of the revolutions executed separately by each of A , B , and C . But as z goes round Γ once, A and B both execute a complete revolution, while the direction of C merely oscillates, so $f(z)$ winds around 0 twice.

It is clear that this result is independent of the circularity of Γ , and that it generalizes to the case of a polynomial $P(z)$ of arbitrary degree: *If a simple loop Γ winds once around m roots of $P(z)$, then $v[P(\Gamma), 0] = m$.*

Roots are simply preimages of 0, and from the geometric viewpoint there is nothing special about this particular image point. Consequently, in future we will look at the preimages of a general point p and we will call these preimages p -points of the mapping.

The *Argument Principle* is a tremendous extension of the above result. Not only does it apply to general analytic mappings but it also contains the converse statement that the winding number tells us the number of preimages:

$$\text{If } f(z) \text{ is analytic inside and on a simple loop } \Gamma, \text{ and } N \text{ is the number of } p\text{-points [counted with their multiplicities] inside } \Gamma, \text{ then } N = v[f(\Gamma), p]. \quad (7.6)$$

The meaning of the expression “counted with their multiplicities” will be explained in the next section.

We wish to stress that this result is only peripherally connected with the conformality that has been so central to all our previous thinking. In fact the Argument Principle is a consequence of a still more general *topological* fact concerning mappings that are merely continuous. Our main effort will therefore be directed towards understanding the general result (due to Poincaré), of which the Argument Principle is merely a special case.

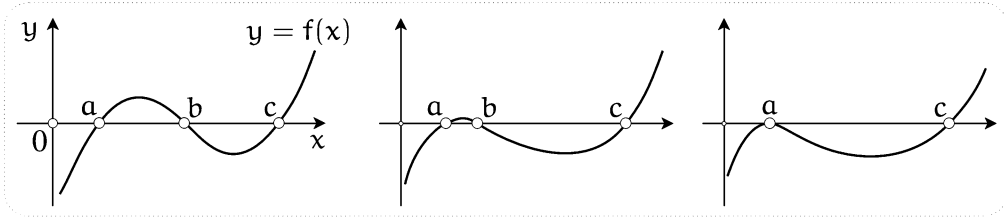
7.4 A Topological Argument Principle*

7.4.1 Counting Preimages Algebraically

Even in the simple case of [7.8] we find that we must be careful how we count preimages. If we move the root b towards the one at a , zero continues to have two preimages inside Γ , while the image of Γ continues to wind round zero twice. However, when b actually *arrives* at a there is apparently only a single 0-point inside Γ (namely a) despite the fact that $f(\Gamma)$ winds round 0 twice. Thus a must now be counted *twice* if (7.6) is to remain true.

Algebraically, the resolution lies in the fact that a is now a “double-root”, the factored form of $f(z) = ABC = A^2C$ containing the *square* of $A = (z - a)$. More generally, if the factorization of a polynomial contains the term A^n then we say that the root a is a 0-point of *algebraic multiplicity* n , and we must count it with this multiplicity in (7.6).

When $n > 1$ there is a further significance to the point a —it is a critical point of the polynomial. In the cubic mapping $f(z) = ABC$, let a , b , and c be real, so that $f(z)$ is real-valued on the real axis. The far left of [7.9] shows the ordinary graph



[7.9] When roots coalesce, the derivative must vanish. *The same applies in $\mathbb{C}$, for if $f'(a) \neq 0$ then an infinitesimal disc centred at a is amplitwisted to an infinitesimal disc centred at 0, so that points close to a cannot map to 0.*

of f in this case. As b moves towards a , the slope at a is forced to decrease, finally vanishing at the moment of b 's arrival.

In general this vanishing of the derivative (now amplitwist) must occur wherever two or more roots of a polynomial coalesce. Look again at [7.9]. If $f'(a) \neq 0$ then the graph is not flat at a and so neighbouring points cannot map to zero; but this is precisely what we insist on when we merge b into a . Essentially the same thing happens when we return to $\mathbb{C}$, for if $f'(a) \neq 0$ then an infinitesimal disc centred at a is amplitwisted to an infinitesimal disc centred at 0, so that points close to a cannot map to 0.

This conclusion can be refined. If the root a of a polynomial $P(z)$ has multiplicity n then P may be factorized as $A^n \Omega(z)$, where $\Omega(a) \neq 0$. It follows by simple calculation [exercise] that the first $(n - 1)$ derivatives of P vanish at a , so that a is a critical point of order at least $(n - 1)$. We shall see in a moment that the order actually is $(n - 1)$.

We next seek to extend the idea of counting preimages “with algebraic multiplicities” to analytic mappings in general. Suppose that a is a p -point of an analytic mapping $f(z)$, i.e., it is a 0-point of $f(z) - p$. What should the algebraic multiplicity of this root be? It is only possible to answer this question because of the remarkable fact that an analytic f can always be represented as a convergent Taylor series in the neighbourhood of a non-singular point. Thus if $\Delta = (z - a)$ is the small complex number from a to a nearby point z , we may write

$$f(z) - p = f(a + \Delta) - f(a) = \frac{f'(a)}{1!} \Delta + \frac{f''(a)}{2!} \Delta^2 + \dots$$

The first nonzero term on the right is the one that dominates the local behaviour of $f(z) - p$ and decides what the multiplicity of a should be. Typically a will not be a critical point [$f'(a) \neq 0$] and so this local behaviour is like Δ to the first power; we say that a is a *simple* root with multiplicity +1.

Now consider the rarer case in which a is a critical point. If the order of the critical point is $(n - 1)$, so that $f^{(n)}$ is the first nonvanishing derivative at a , then the dominant first term is proportional to Δ^n , and we correspondingly define the

*algebraic multiplicity*² of a to be n . The analogy between this definition and that for polynomials may be brought to the fore by setting

$$\Omega(z) = \frac{f^{(n)}(a)}{n!} + \frac{f^{(n+1)}(a)}{(n+1)!}\Delta + \frac{f^{(n+2)}(a)}{(n+2)!}\Delta^2 + \dots,$$

where $f^{(n)}(a)$ is the first nonvanishing derivative. The previous equation can now be written in “factorized” form as

$$f(z) - p = \Omega(z) \Delta^n, \quad (7.7)$$

where $\Omega(a) \neq 0$. From this point of view, the only difference between a general analytic mapping and a polynomial is that the latter has a single, “once and for all” factorization, while the former generally requires a different factorization of type (7.7) in the neighbourhood of each p -point.

7.4.2 Counting Preimages Geometrically

Recall that we wish to explain (7.6) as a special case of a more general result dealing with mappings that are merely continuous. But since the very notion of algebraic multiplicity is meaningless for such general mappings, how can we even frame a proposition of type (7.6)?

What is needed is a geometric way of counting preimages that will agree with the previous algebraic definition if we specialize to analytic mappings. To discover the appropriate definition we should therefore return to analytic mappings and ask, “What is the *geometric fingerprint* of a p -point of given algebraic multiplicity?”

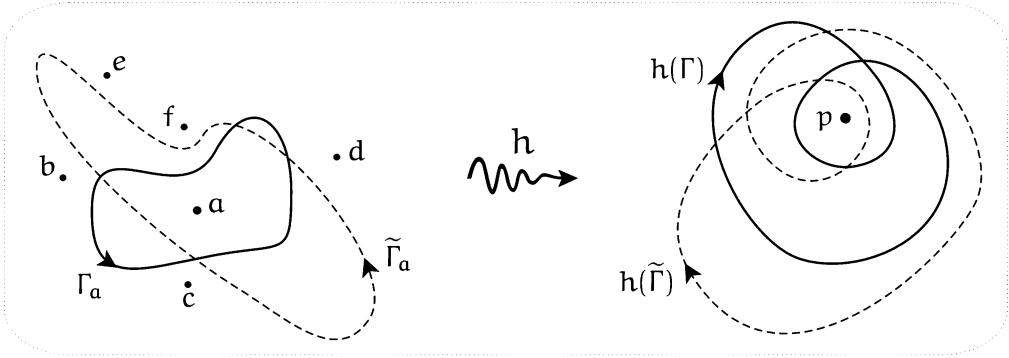
Consider the effect of an analytic f on an infinitesimal circle C_a centred at a *simple* p -point a . Since $f'(a) \neq 0$, C_a is amplitwisted to an infinitesimal circle centred at p . We see that the winding number (+1) of this image round p is the same as the algebraic multiplicity of a . In fact, quite generally, if the algebraic multiplicity is n then the winding number of the image will also be n . This is the sought-after geometric fingerprint.

To verify this statement, remember that the local behaviour of f near to a is given by (7.7):

$$f(z) = f(a + \Delta) = p + \Omega(z) \Delta^n,$$

with $\Omega(a) \neq 0$. Thus the basic explanation is that as Δ revolves round C_a once, Δ^n rotates n -times as fast, and therefore $f(z)$ completes n revolutions round p . If

² Also known as *order* or *valence*.



[7.10] Topological Multiplicity as generalization of Algebraic Multiplicity. On the left are p -points of the continuous mapping h , i.e., points that are mapped to p by h , and Γ_a is a loop around a (and only a). If we continuously deform Γ_a into another such loop $\tilde{\Gamma}_a$ without crossing a (or any other p -point) then $h(\Gamma_a)$ will continuously deform, but its winding number around p will not change. Thus $\nu(a) \equiv \nu[h(\Gamma_a), p]$ is really a topological invariant associated with the point a , and it is called its topological multiplicity. If we specialize h to be a complex polynomial, then the topological multiplicity is equal to the algebraic multiplicity of a as a repeated root of $h(z) = p$.

Ω were a constant then this argument would be beyond reproach, and we can now do a little calculation to show that a variable Ω does not disturb the conclusion:

$$\begin{aligned} \nu[f(C_a), p] &= \nu[f(C_a) - p, 0] = \nu[\Delta^n \Omega(C_a), 0] \\ &= n \nu[\Delta, 0] + \nu[\Omega(C_a), 0]. \end{aligned} \quad (7.8)$$

As we shrink C_a down towards a , $\Omega(C_a)$ will shrink down towards $\Omega(a)$, but since $\Omega(a) \neq 0$ this implies that the image of a sufficiently small C_a will not wind round 0: $\nu[\Omega(C_a), 0] = 0$. Since $\nu[\Delta, 0] = 1$, we conclude that $\nu[f(C_a), p] = n$, as claimed.

We may now broaden our horizon and use the above idea to define “multiplicity” for a mapping $h(z)$ that is merely continuous. Let Γ_a be any simple loop round a that does not contain other p -points. Figure [7.10] shows such a loop as well as some other p -points b, c , etc. If we continuously deform Γ_a into another such loop $\tilde{\Gamma}_a$ without crossing a (or any other p -point) then $h(\Gamma_a)$ will continuously deform into $h(\tilde{\Gamma}_a)$ without ever crossing p , and so

$$\nu[h(\tilde{\Gamma}_a), p] = \nu[h(\Gamma_a), p].$$

Thus, without specifying Γ_a further, we may unambiguously define the *topological multiplicity*³ of a to be

$$\nu(a) \equiv \nu[h(\Gamma_a), p].$$

³ Also known as the *local degree* of h at a .

In the case of the mapping in [7.10] we see that $v(a) = -2$. If h happened to be analytic then a would also possess an algebraic multiplicity n , but by deforming Γ_a into the infinitesimal circle C_a , we find that the two kinds of multiplicity must agree: $v(a) = n$.

7.4.3 What's Topologically Special About Analytic Functions?

From the geometric point of view, conformal (analytic) mappings are infinitely richer in structure than mappings that are merely continuous. However, from the point of view of topological multiplicity there are only a few distinctions, the following being one of the most striking:

$v(a)$ is always positive for analytic functions, while it can be negative for nonanalytic functions.

For example, the mapping in [7.10] cannot possibly be analytic. The positivity of $v(a)$ for analytic functions has already been established, so we need only look more closely at the possibility of negative multiplicities for nonanalytic functions.

Since general continuous mappings can actually behave in rather wild ways, let us restrict ourselves to nonanalytic mappings that are at least differentiable in the real sense. For example, consider $h(z) = \bar{z}$. The unique preimage of p is $a = \bar{p}$, and any simple loop Γ_a round this point is reflected by h into a loop that goes once round p in the opposite direction. Thus $v(a) = -1$.

More generally, recall [see p. 236] that the local effect of such a mapping at a p -point a is (after translation to p) a stretch by some factor ξ_a in one direction, another stretch by some factor η_a in the perpendicular direction, and finally a rotation through some angle ϕ_a . For example, conjugation has (taking the first expansion to be horizontal) $\xi_a = +1$, $\eta_a = -1$, $\phi_a = 0$. Of course these values of ξ_a , η_a , and ϕ_a are only independent of a because $h(x + iy) = x - iy$ depends linearly on x and y ; most mappings have values that *do* depend on the point a .

An infinitesimal circle C_a centred at a is generally distorted into an infinitesimal ellipse E_p centred at p , and if the two expansion factors have the same sign then the mapping *preserves orientation* so that E_p circulates in the same sense as C_a and $v(a) = +1$. However, if ξ_a and η_a have opposite signs then the mapping is *orientation reversing*: it turns C_a inside out, so that E_p goes round p in the opposite direction and $v(a) = -1$. Our previous example of conjugation was of this type. In summary, we have

$$v(a) = \text{the sign of } (\xi_a \eta_a).$$

The local linear transformation at a is encoded by the Jacobian matrix $J(a)$, and we can use its determinant $\det[J(a)]$ to give a more practical formula for the topological multiplicity. We know from linear algebra that the determinant of a constant 2×2 matrix is the factor (including a sign for orientation) by which the area of a

figure is expanded. Likewise, $\det[J(a)]$ measures the *local* expansion factor for area at a , and this is just $(\xi_a \eta_a)$. Thus

$$v(a) = \text{the sign of } \det[J(a)]. \quad (7.9)$$

Of course this formula is vacuous if $\det[J(a)] = 0$. Geometrically this means that the transformation is locally crushing at a ; just as for analytic mappings, such a place is called a *critical point*. However, while the local crushing at a critical point of an analytic mapping is perfectly symmetrical in all directions, this is not true of the more general mappings presently under consideration. For example, if $f(x + iy) = x - iy^3$ then [exercise] $\det[J] = -3y^2$, so although f leaves the horizontal separation of points alone, all the points on the real axis are critical points as a result of crushing in the vertical direction.

This example also serves to illustrate another difference:

The critical points of an analytic mapping can be distinguished purely on the basis of topological multiplicity; those of a nonanalytic mapping cannot.

For analytic functions we have seen that $v(a) = +1$ if and only if a is not critical. In the nonanalytic case $v(a) = \pm 1$ if a is not critical, but it is also possible for a *critical point* to have one of these multiplicities. Indeed, you can check [exercise] that the above example yields $v(a) = -1$ for noncritical and critical points alike.

One final difference:

$v(a)$ is never zero for analytic mappings, but it can vanish for nonanalytic mappings.

In the next section we will provide a simple example of a nonanalytic mapping possessing such p -points of vanishing topological multiplicity. Can you think of an example for yourself?

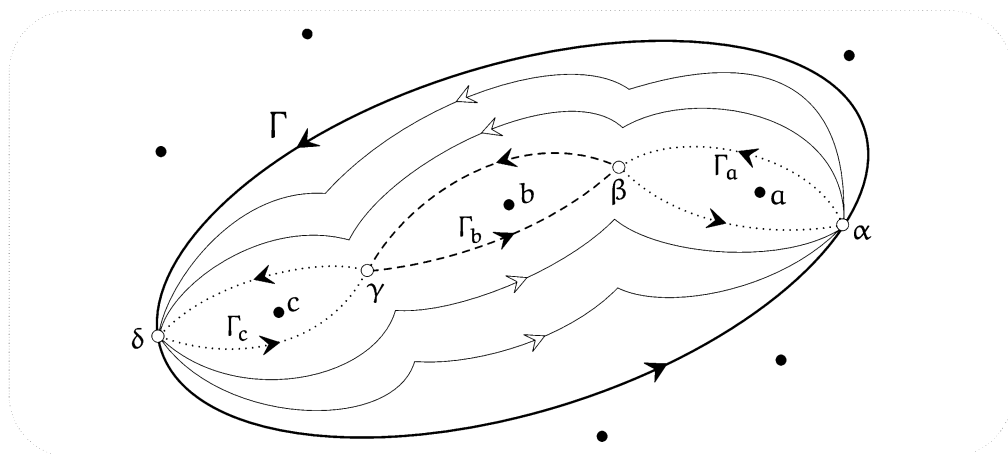
7.4.4 A Topological Argument Principle

Let Γ be a simple loop, and let $h(z)$ be a continuous mapping such that only a finite number of its p -points lie inside Γ . We will show that

$$\text{The total number of } p\text{-points inside } \Gamma \text{ (counted with their topological multiplicities) is equal to the winding number of } h(\Gamma) \text{ round } p. \quad (7.10)$$

If h is analytic this reduces to (7.6). The rest of the chapter will be devoted to mining and extending this simple yet profound result⁴.

⁴ When interpreted in terms of vector fields (as we shall do in Chapter 10) this is the key to a very surprising and beautiful fact called the *Poincaré–Hopf Theorem*.



[7.11] The winding number $\nu[h(\Gamma), p]$ equals the sum of the topological multiplicities of the p -points inside Γ . For this winding number will not change value if we deform Γ as shown into a sum of loops around the individual p -points within.

Before explaining (7.10), let us describe one of its immediate consequences. As in [7.2], $h(\Gamma)$ will generally partition the plane into a number of sets, and the above result then says that every point in D_j has the same number of preimages lying inside Γ , namely, ν_j . For example, if $h(\Gamma)$ is a simple loop then it partitions the plane into just two sets, namely, its interior and its exterior. If p is in the interior then the result says that the total number of p -points inside Γ is 1. But if h is analytic then these p -points must have strictly positive multiplicities, and so there is exactly one preimage for each point inside $h(\Gamma)$. In other words, we have shown that

If an analytic function h maps Γ onto $h(\Gamma)$ in one-to-one fashion, then it also maps the interior of Γ onto the interior of $h(\Gamma)$ in one-to-one fashion.

This is *Darboux's Theorem*.

To explain (7.10), consider [7.11]. This shows three p -points a , b , and c lying inside Γ while others lie scattered outside. The essential idea is that we can gradually deform Γ (as shown) into the doubly-pinched loop $\alpha\beta\gamma\delta\gamma\beta\alpha$, which we will call $\tilde{\Gamma}$. Since no p -points were crossed in the deformation process, $h(\tilde{\Gamma})$ will wind round p the same number of times as $h(\Gamma)$. The rest is almost obvious: $\tilde{\Gamma}$ is made up of $\Gamma_a = \alpha\beta\alpha$, $\Gamma_b = \beta\gamma\beta$, $\Gamma_c = \gamma\delta\gamma$, and the winding numbers of their images round p are, by definition, the topological multiplicities of a , b , and c .

We will spell this out in perhaps unnecessary detail. Let K be a path that is not necessarily closed, and define $\mathcal{R}(K)$ to be the net rotation of $h(z)$ round p as z traverses K . For example, if K is closed then $\mathcal{R}(K) = 2\pi\nu[h(K), p]$. Then,

$$\begin{aligned}
2\pi v[h(\Gamma), p] &= 2\pi v[h(\tilde{\Gamma}), p] \\
&= \mathcal{R}(\alpha\beta\gamma\delta\gamma\beta\alpha) \\
&= \mathcal{R}(\alpha\beta) + \mathcal{R}(\beta\gamma) + \mathcal{R}(\gamma\delta) + \mathcal{R}(\delta\gamma) + \mathcal{R}(\gamma\beta) + \mathcal{R}(\beta\alpha) \\
&= \mathcal{R}(\alpha\beta\alpha) + \mathcal{R}(\beta\gamma\beta) + \mathcal{R}(\gamma\delta\gamma) \\
&= \mathcal{R}(\Gamma_a) + \mathcal{R}(\Gamma_b) + \mathcal{R}(\Gamma_c) \\
&= 2\pi[v(a) + v(b) + v(c)].
\end{aligned}$$

Clearly this idea extends to any number of p -points a_1, a_2 , etc. lying inside Γ :

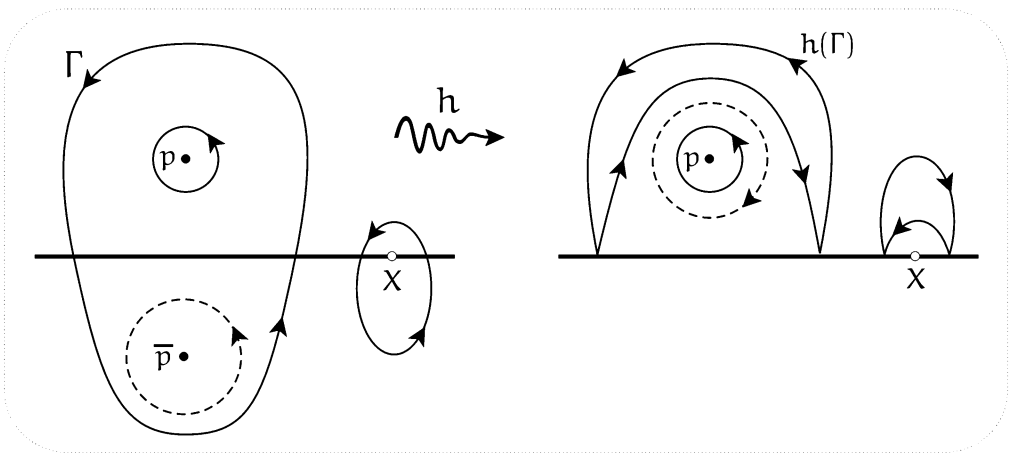
$$v[h(\Gamma), p] = \sum_{a_j \text{ inside } \Gamma} v(a_j). \quad (7.11)$$

7.4.5 Two Examples

Let us immediately illustrate the result with a concrete mapping:

$$h(x + iy) = x + i|y|.$$

In terms of our pastry analogy [p. 234] this corresponds to making a crease along the real axis and folding the bottom half of the plane onto the top half; in fact, since no stretching is required in this example, folding a simple piece of paper will do! If $\text{Im}(p) > 0$ then p has two preimages, $a_1 = p$ and $a_2 = \bar{p}$. Figure [7.12] shows that $v(a_1) = +1$ and $v(a_2) = -1$, and in accord with (7.11), it also shows that if Γ contains a_1 and a_2 then $v[h(\Gamma), p] = 0$.



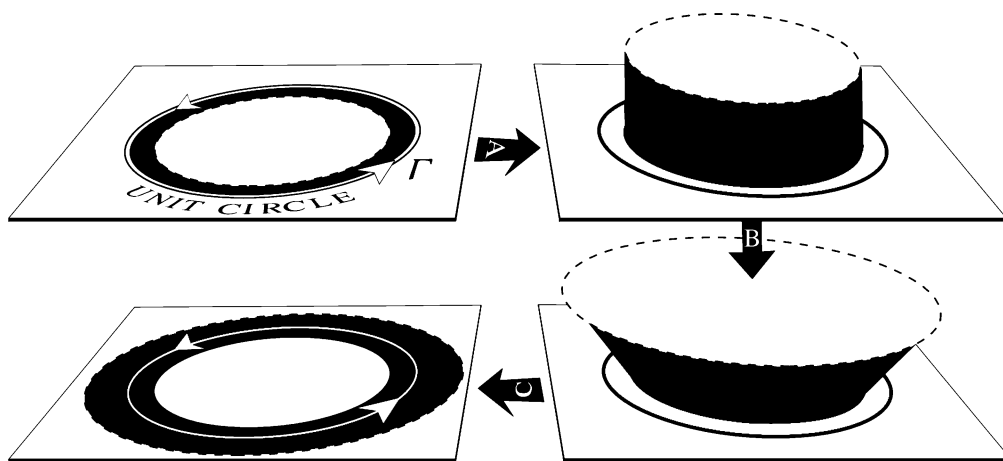
[7.12] Folding a piece of paper across the x -axis is a continuous mapping h , explicitly given by $h(x + iy) = x + i|y|$. If $\text{Im}(p) > 0$, as shown, then p has two preimages, $a_1 = p$ and $a_2 = \bar{p}$, so $v(a_1) = +1$ and $v(a_2) = -1$. Therefore, (7.11) predicts that $v[h(\Gamma), p] = v(a_1) + v(a_2) = (+1) + (-1) = 0$, which is indeed true!

In general, note that $v[h(\Gamma), p] = 0$ merely implies that *either* there are no preimages inside Γ *or* the preimages have cancelling topological multiplicities, as above. However, if f is analytic and $v[f(\Gamma), p] = 0$ then the conclusion is quite definite: there are no preimages inside Γ . Later we shall return to this important point.

Returning to the example, observe that if $p = X$ is real then there is only one preimage, namely X , and $v(X) = 0$. We can look at this in a nice way: as we move p towards X , the two preimages α_1 and α_2 also move towards X , and when they finally coalesce at X their opposite multiplicities annihilate. As we previously pointed out, such points of vanishing multiplicity can *only* exist for nonanalytic mappings.

Figure [7.13] shows a second more elaborate example, in which we subject a unit disc of pastry to a three-stage transformation H that leaves the boundary Γ (the unit circle) fixed: $H(\Gamma) = \Gamma$. Here are the three stages: (A) form a “hat” by lifting up the part of the disc lying inside the dashed circle, some of the pastry dough outside the dashed circle being stretched to form the side of the cylinder. (B) radially stretch the disc forming the top of this “hat” till its radius is greater than one. (C) press down flat, i.e., project each point vertically down onto the plane.

If we pick a point p from the image set [bottom left] then the number of preimages (counted naively) lying in the original disc is the number of layers of pastry lying over p . In the final picture [bottom left] we have used the degree of shading to indicate the number of these layers: *one* over the lightly shaded inner disc; *two*



[7.13] (A) Form a “hat” by lifting up the part of the disc lying inside the dashed circle, some of the pastry dough outside the dashed circle being stretched to form the side of the cylinder. (B) Radially stretch the disc forming the top of this “hat” till its radius is greater than one. (C) Press down flat, i.e., project each point vertically down onto the plane. The final number of layers [bottom left] is indicated by the shading: one layer over the lightly shaded inner disc; two over the darkly shaded outer ring; and three over the black ring. You can now geometrically verify that (7.11) works in all three regions.

over the darkly shaded outer ring; *three* over the black ring. Make sure you can see this.

We can now check (7.11). For example, if p lies in the darkly shaded outer ring, $\nu[H(\Gamma), p] = \nu[\Gamma, p] = 0$ and so the multiplicities of the preimages of such points should sum to zero. By following the effect of the transformation on little loops round each of the two preimages, we confirm this prediction: one preimage has multiplicity $+1$ while the other has multiplicity -1 .

Check for yourself that (7.11) continues to work if p instead lies in the inner disc or in the black ring.

7.5 Rouché's Theorem

7.5.1 The Result

Imagine walking a dog round and round a tree in a park, both you and the dog finally returning to your starting points. Further imagine that the dog is on one of those leashes of variable length, similar to a spring-loaded tape measure. On one such walk you keep the leash short, so that the dog stays at your heel. It is then clear that the dog is forced to walk round the tree the same number of times that you do. On another walk, though, you decide to let out the leash somewhat so that the dog may scamper about, perhaps even running circles around you. Nevertheless, provided that you *keep the leash short enough so that the dog cannot reach the tree*, then again the dog must circle the tree the same number of times as you.

Let the tree be the origin of $\mathbb{C}$, and let your walk be the image path traced by $f(z)$ as z traverses a simple loop Γ . Also, let the complex number from you to the dog be $g(z)$, so that the dog's position is $f(z) + g(z)$. The requirement that the leash not stretch to the tree is therefore

$$|g(z)| < |f(z)| \text{ on } \Gamma.$$

Under these circumstances, the previous paragraph states that

$$\nu[(f + g)(\Gamma), 0] = \nu[f(\Gamma), 0].$$

But the Argument Principle then informs us that

If $|g(z)| < |f(z)|$ on Γ , then $(f + g)$ must have the same number of zeros inside Γ as f .

This is *Rouché's Theorem*.

Note that while $|g(z)| < |f(z)|$ is a sufficient condition for $(f + g)$ to have the same number of roots as f , it is not a necessary condition. For example, consider $g(z) = 2f(z)$.

7.5.2 The Fundamental Theorem of Algebra

A classic illustration of Rouché's Theorem is the Fundamental Theorem of Algebra, which states that a polynomial

$$P(z) = z^n + A z^{n-1} + B z^{n-2} + \cdots + E$$

of degree n always has n roots. The basic explanation is simple: if $|z|$ is large, the first term dominates the behaviour of $P(z)$ and the image of a sufficiently large origin-centred circle C will therefore wind n times round 0; the Argument Principle then says that $P(z)$ must have n roots inside C .

Rouché's Theorem merely allows us to make the above idea more precise. Let $f(z) = z^n$ be the first term of $P(z)$ and let $g(z)$ be the sum of all the rest, so that $f + g = P$. Now let C be the circle $|z| = 1 + |A| + |B| + \cdots + |E|$. Using the fact that $|z| > 1$ on C , it is not hard to show [exercise] that $|g(z)| < |f(z)|$ on C , and since f has n roots inside C (all at the origin), Rouché says that P must too.

Notice that we have not only confirmed the existence of the n roots, but have also narrowed down their location: they must all lie inside C . In the exercises you will see how Rouché's Theorem can often be used to obtain more precise information on the location of the roots of an equation.

7.5.3 Brouwer's Fixed Point Theorem*

Sprinkle talcum powder on a cup of coffee and give it a stir. The little white specks will swirl around and eventually come to rest, the speck that was originally at z finally ending up at $g(z)$. If we stir it in a nice symmetrical way then the speck in the centre will remain motionless and its final position will be identical with its starting position. Such a place, for which $g(z) = z$, is called a *fixed point* of g .

Now stir the coffee in a really complicated way and let it again come to rest. Incredible as it may seem, at least one speck will have ended up exactly where it started! This is an example of *Brouwer's Fixed Point Theorem*, which asserts that any continuous mapping of the disc to itself will have a fixed point. Exercise 15 shows this to be true, but for the moment we wish to demonstrate a slightly different result: there must be a fixed point if the disc is mapped into its *interior* and there are at most a finite number of fixed points.

Let the disc D be $|z| \leq 1$; the condition that g map D into the interior of D is then $|g(z)| < 1$ for all z in D . Let $m(z)$ be the movement of z under the mapping, i.e., the connecting complex number from z to its destination $g(z)$:

$$m(z) = g(z) - z.$$

A fixed point then corresponds to no movement: $m(z) = 0$. Now let $f(z) = -z$. On the boundary of D (the unit circle) we have

$$|g(z)| < 1 = |f(z)|$$

and so Rouché's Theorem says that $m(z) = g(z) + f(z)$ has the same number of roots inside D as f has, namely, one.

If g is merely continuous then there can actually be several fixed points, some of which will necessarily have negative multiplicities, while if g is analytic then there can literally only be one.

7.6 Maxima and Minima

7.6.1 Maximum-Modulus Theorem

Take another look at the nonanalytic mapping H of [7.13], and note how the image of the disc “spills over” the image of its boundary: points inside Γ end up in the darkly shaded ring *outside* $H(\Gamma)$. The central observation of this section is that such spilling over is quite impossible in the case of an analytic mapping:

If f is analytic inside and on a simple loop Γ then no point outside $f(\Gamma)$ can have a preimage inside Γ . (7.12)

Let's see why. The Argument Principle tells us that the sum of the multiplicities $v(a_j)$ of the p -points inside Γ is $v[f(\Gamma), p]$, but if p is outside $f(\Gamma)$ then (by definition) this is zero. Since $v(a_j)$ is strictly positive for an analytic function, we conclude that points outside $f(\Gamma)$ have no preimages inside Γ . On the other hand, if p lies inside $f(\Gamma)$ then $v[f(\Gamma), p] \neq 0$, and so there must be at least one preimage inside Γ . [Unlike (7.12), this is also true of nonanalytic mappings.]

Figure [7.14] illustrates an analytic f sending the shaded interior of Γ strictly to the shaded interior of $f(\Gamma)$. Since $f(\Gamma)$ winds round the darker region twice, its points have two preimages in Γ ; we can think of this as arising from the overlap of two lightly shaded regions, one preimage per lightly shaded point.

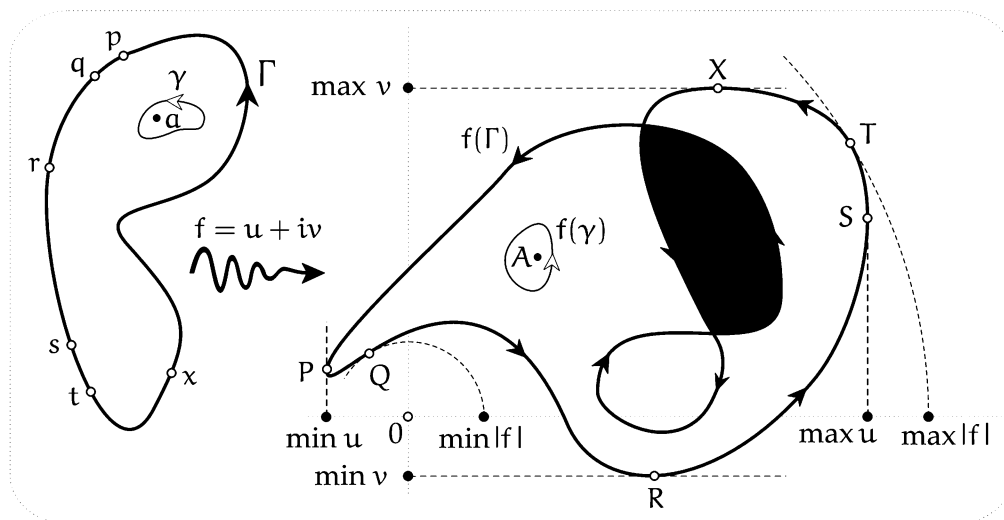
One aspect of the “overspill” produced by H in [7.13] is that the points z which end up furthest from the origin (i.e., for which the modulus $|H(z)|$ is maximum) lie inside Γ . Conversely, the absence of overspill for an analytic f means that

The maximum of $|f(z)|$ on a region where f is analytic is always achieved by points on the boundary, never ones inside.

This is called the *Maximum-Modulus Theorem* and it is illustrated in [7.14]: the maximum of $|f(z)|$ is $|T| = |f(t)|$, where t lies on Γ .

The only exception to this result is the trivial analytic mapping $z \mapsto \text{const.}$, which sends every point to a single image point. To put this more positively, if we know that $|f|$ achieves its maximum at an interior point then $f(z) = \text{const.}$

As an elementary example, consider this problem. Let $F(z)$ be the product of the distances from z to the vertices a, b, c, d of a square. If z lies inside or on the edge of the square, where does the maximum value of F occur? It is certainly tempting



[7.14] Maximum-Modulus Theorem: *The maximum of $|f(z)|$ on a region where f is analytic is always achieved by points on the boundary, never ones inside. For the Argument Principle tells us that the sum of the multiplicities $\nu(a_j)$ of the p -points inside Γ is $\nu[f(\Gamma), p]$, but if p is outside $f(\Gamma)$ then (by definition) this is zero. Since $\nu(a_j)$ is strictly positive for an analytic function, we conclude that points outside $f(\Gamma)$ have no preimages inside Γ . On the other hand, if p lies inside $f(\Gamma)$ then $\nu[f(\Gamma), p] \neq 0$, and so there must be at least one preimage inside Γ .*

to guess that the maximum will occur at the centre of the square, but this is wrong. Since $F(z) = |(z - a)(z - b)(z - c)(z - d)|$ is the modulus of an analytic mapping, the maximum must in fact occur somewhere on the edge of the square. The exact location can now be found [exercise] using nothing more than ordinary calculus.

Returning to matters of theory, recall from Chapter 2 that the “modular surface” of f is the surface obtained by lifting each point z vertically to a height $|f(z)|$ above the complex plane. If we look at the portion of this surface lying above Γ and its interior, the result says that the highest point always lies on the edge, never inside.

Although the absolute maximum of the height always occurs on the edge, could there perhaps be a *local* maximum of $|f|$ at an interior point a , so that the surface would have a peak above a ? No! For if we cut out the piece of the surface lying above the interior of any small loop γ round a , the highest point will fail to lie on the edge. Thus a modular surface has no peaks. Further aspects of the modular surface are investigated in the exercises.

This absence of local maxima is re-explained in [7.14]. The Argument Principle says that since γ contains a , $f(\gamma)$ must wind round $A = f(a)$ at least once. This makes it clear that there are always points on γ which have images lying further from the origin than A . More formally (cf. [7.4]), any ray emanating from A must intersect $f(\gamma)$, and by choosing the ray to point directly away from the origin, the intersection point is guaranteed to lie further from the origin than A . Thus $|f|$ cannot have a local maximum.

7.6.2 Related Results

As [7.14] illustrates, the Maximum-Modulus result is only one of several that follow from (7.12). For example, unless there is a 0-point inside Γ , at which $|f(z)| = 0$, the point Q closest to the origin (for which $|f(z)|$ is minimum) must also be the image of a point q lying on the boundary Γ . Naturally enough, this is called the *Minimum-Modulus Theorem*.

Thus if we cut out the piece of the modular surface lying above Γ and its interior, the lowest point will always lie on the edge, unless, that is, the surface actually hits the complex plane at an interior 0-point of f . By the same token, there can be no pits in the surface [local minima of $|f|$] except at 0-points.

As before, the only exception to all this is the mapping $z \mapsto \text{const.}$, for which every point yields the smallest (and only) value of $|f|$. Thus if we know that $|f|$ achieves a positive minimum at an interior point then $f(z) = \text{const.}$

If $f = u + iv$ is analytic then [cf. Ex. 2, p. 293] u and v are automatically “harmonic”. As we shall see in Chapter 11, this means that these functions are intimately connected with numerous physical phenomena: heat flow, electrostatics, hydrodynamics, to name but a few. It is therefore of significance that [7.14] shows that *u and v are also subject to the principle that their maxima and (nonzero) minima can only occur on Γ , never inside Γ* . As before, if a maximum or minimum occurs at an interior point, the harmonic function must be constant.

7.7 The Schwarz–Pick Lemma*

7.7.1 Schwarz’s Lemma

Thinking of the unit disc as the Beltrami–Poincaré model for non-Euclidean geometry, we saw in Chapter 6 that a special role was played by the Möbius transformations of the form

$$M_a^\phi(z) = e^{i\phi} \left(\frac{z - a}{\bar{a}z - 1} \right) = e^{i\phi} M_a(z), \quad (7.13)$$

where a lies inside the disc. These one-to-one mappings of the disc to itself act as rigid motions, for they preserve non-Euclidean distance.

Apart from a digression on Liouville’s Theorem, this section continues the work (begun in Chapter 6) of exhibiting the beautiful pre-existing harmony that exists between non-Euclidean geometry and the theory of conformal mappings. Our first new piece of evidence that these two disciplines somehow “know” about each other is the following:

$$\textit{Rigid motions of the hyperbolic plane are the \textbf{only} one-to-one analytic mappings of the disc to itself.} \quad (7.14)$$

There are of course many other kinds of analytic mapping of the disc to itself, but according to (7.14) they must all fail to be one-to-one. For example, $z \mapsto z^3$ maps the disc to itself, but it is *three-to-one*.

Observe that this result establishes a claim we previously made [see p. 204] in connection with Riemann's Mapping Theorem. There we explained that there are as many mappings of one region to another as there are automorphisms of the disc. We already knew that these automorphisms included 3-parameter's worth of Möbius mappings, and (7.14) now tells us that there are no more.

To verify (7.14) we will first establish a lemma (of great interest in itself) due to Schwarz:

If an analytic mapping of the disc to itself leaves the centre fixed, then either every interior point moves nearer to the centre, or else the transformation is a simple rotation.

The example $f(z) = z^2$ shows that the mapping need not be a rotation in order for boundary points to keep their distance from the centre. However, at an interior point we have $|z| < 1$, and so $|f(z)| = |z|^2 < |z|$, in accord with the result.

Let f be any analytic mapping of the disc to itself leaving the centre fixed, so that $|f(z)| \leq 1$ on the disc, and $f(0) = 0$. We wish to show that either $|f(z)| < |z|$ at interior points, or else $f(z) = e^{i\phi} z$. To this end, consider the ratio F of image to preimage:

$$F(z) \equiv \frac{f(z)}{z}.$$

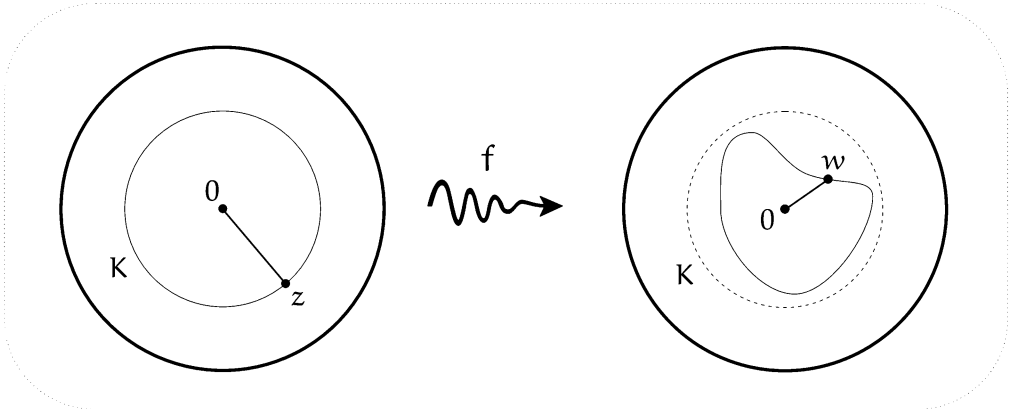
At first sight this may look undefined at 0, but a moment's thought shows that as z approaches the origin, $F(z)$ approaches $f'(0)$.

From the previous section we know that the maximum modulus of an analytic function on the disc can only occur at an interior point if the function is constant, otherwise it's on the boundary circle $|z| = 1$. Thus if p is an interior point and z varies over the unit circle C , then

$$|F(p)| \leq (\max |F(z)| \text{ on } C) = (\max |f(z)| \text{ on } C) \leq 1.$$

Thus it is certainly true that no interior point can end up *further* from the centre. But if even a single interior point q remains at the *same* distance from the centre then $|F(q)| = 1$, which means that F has achieved its maximum modulus at an interior point. In this case F must map the entire disc to a single point of unit modulus, say $e^{i\phi}$, so that $f(z) = e^{i\phi} z$ is a rotation. Done.

The result is illustrated in [7.15]. If f is not a rotation then every point z on a circle such as K is mapped to a point $w = f(z)$ lying strictly inside K , and the shaded region is compressed as shown. If we shrink K down towards the origin then f will amplitwist it to another infinitesimal circle centred at the origin, but having a



[7.15] Schwarz's Lemma: *If an analytic mapping $f(z)$ of the disc to itself leaves the centre fixed, then either every interior point moves nearer to the centre, or else the transformation is a simple rotation. As explained in the text, this follows from applying the Maximum-Modulus Theorem to $F(z) \equiv f(z)/z$.*

smaller radius. Thus the amplification of f at the origin must be less than one. We can only have $|f'(0)|$ equal to one in the case of a rotation.

We can now return to (7.14). As in Schwarz's Lemma, first suppose that the mapping f leaves the centre fixed, but now take f to be *one-to-one*, so that it has a well-defined analytic inverse f^{-1} which also maps the disc to itself and leaves the centre fixed. By Schwarz's Lemma, f sends an interior point p to a point q that is no further from centre than p . But f^{-1} is *also* subject to Schwarz's Lemma, and so $p = f^{-1}(q)$ must be no further from the centre than q . These two statements are only compatible if $|q| = |f(p)| = |p|$. Thus f must be a rotation, which is indeed a rigid motion of type (7.13).

Finally, suppose that the one-to-one mapping f does not leave the centre fixed, but instead sends it to c . We can now compose f with the rigid motion M_c which sends c back to 0. We thereby obtain a one-to-one mapping $(M_c \circ f)$ of the disc to itself which does leave the centre fixed, and which must therefore be a rotation M_0^ϕ . But this means [exercise] that

$$f = M_c \circ M_0^\phi$$

is the composition of two rigid motions, and so is itself a rigid motion. Done.

7.7.2 Liouville's Theorem

The constant mapping $f(z) = c$ crushes the entire plane down to the single image point c . We now ask whether it is possible for an analytic mapping to compress

the entire plane down to a region lying inside a finite circle, *without* going to the extreme of completely crushing it to a point.

If we merely demand that the mapping be continuous then this can happen. For example,

$$h(z) = \frac{z}{1 + |z|}$$

maps the entire plane to the unit disc. Returning to analytic mappings, we notice that complex inversion, $z \mapsto w = (1/z)$, manages to conformally compress the infinite region lying outside the unit circle of the z -plane into the unit disc of the w -plane. This looks quite hopeful: of the original plane only a puny unit disc remains to be mapped.

To think like this is to completely forget the rigidity of analytic mappings. Having decided to use complex inversion to map the region outside the unit circle, we cannot change the rules when it comes to mapping the remaining disc: the mapping $z \mapsto (1/z)$ acting in the exterior can only be analytically continued to the interior in one way, namely as $z \mapsto (1/z)$. The requirement of analyticity thereby forces the “puny” disc to explode, producing an image of infinite size.

We will now show that

An analytic mapping cannot compress the entire plane into a region lying inside a disc of finite radius without crushing it all the way down to a point.

This is *Liouville's Theorem*. To understand this we must generalize Schwarz's Lemma slightly. Suppose that an analytic function $w = f(z)$ leaves the origin fixed and compresses the disc $|z| \leq N$ to a region lying inside the disc $|w| \leq M$. By the same reasoning as before, we find that if p lies inside the original disc (boundary circle K) then

$$|f(p)| \leq [\max |f(z)| \text{ on } K] = \max \left(\frac{|f(z)|}{N} \right) \text{ on } K \leq \frac{M}{N}.$$

Hence,

$$|f(p)| \leq \frac{M|p|}{N}.$$

But if f compresses the *whole* plane to a region lying inside the disc of radius M , then the above result will continue to hold true no matter how large we make N . Therefore $f(p) = 0$ for all p , and we are done.

Finally, if f does not leave the origin fixed, but instead sends it to c , we may apply the previous argument to the function $[f(z) - c]$. This is the composition of f with the translation which sends c back to 0. Since the image of the plane under f lies inside the disc $|w| \leq M$, the translation of $-c$ will produce a region lying inside the disc $|w| \leq 2M$. The previous inequality then becomes

$$|f(p) - c| \leq \frac{2M|p|}{N}.$$

Once again letting N tend to infinity, we conclude that $f(p) = c$ for all p . Done.

7.7.3 Pick's Result

We now turn to a second, rather beautiful piece of evidence that non-Euclidean geometry is intimately connected with the theory of conformal mapping. Reconsider [7.15]. Schwarz's Lemma informs us that (with the exception of rotations) the distance between interior points and the origin is decreased. This result has two blemishes, both related to an exaggerated emphasis on the origin: (i) we require that the mapping leave the origin fixed; (ii) only distances from the origin are shown to decrease.

Consider (ii) first, and for the moment let us simply put up with (i) by continuing to assume that our analytic mapping leaves the origin fixed. Although we did not demonstrate it, perhaps a more symmetrical result holds true—with the exception of a rotation, *will the mapping automatically decrease the distance between any pair of interior points?*

Sadly, no. Consider the effect of $f(z) = z^2$ (which leaves the origin fixed) on the two interior points $a = (3/4)$ and $b = (1/2)$. The original separation is $|a - b| = 0.25$, while the separation of the images is $|f(a) - f(b)| = 0.3125$. The distance between the pair of points has *increased*.

But now consider the effect of exactly the same mapping on exactly the same two points from the point of view of the *Beltrami–Poincaré*⁵. When *they* measure the distance between a and b , it is found [see (6.46), p. 366] to equal $\mathcal{H}\{a, b\} = 0.8473$, while the separation of the images is $\mathcal{H}\{f(a), f(b)\} = 0.7621$. *The hyperbolic distance has decreased!* Choose any other pair of points for yourself and examine the effect of $z \mapsto z^2$ on their hyperbolic separation.

Pick's splendid discovery⁶ was that even if we drop the requirement that the origin be a fixed point, this decrease in *hyperbolic* distance is a universal phenomenon:

*Unless an analytic mapping of the disc to itself is a hyperbolic rigid motion,
the hyperbolic separation of every pair of interior points decreases.* (7.15)

⁵ Recall from Chapter 6 that this is the race of beings who inhabit the Beltrami–Poincaré model of the hyperbolic plane.

⁶ Georg Alexander Pick [1859–1942] is perhaps best known for his wonderfully simple but completely unexpected *Pick's Theorem*, which determines the area $\mathcal{A}$ of a lattice polygon in terms of the number of lattice points Inside it (I), and on its Boundary (B), namely, $\mathcal{A} = I + (B/2) - 1$. However, his greatest contribution to science may have been the fact that he fought to appoint Einstein at the German University of Prague in 1911, became close friends with him there—playing violin together in a musical quartet—and finally introduced Einstein to the tensor calculus of Ricci and Levi-Civita, without which Einstein could not have formulated General Relativity in 1915! [This is documented in Goodstein (2018, p. 90), though, strangely, Einstein himself only seems to have publicly credited his friend Marcel Grossmann for this critical piece of guidance.] Pick died at the hands of the Nazis at Theresienstadt concentration camp on the 26th of July, 1942, at the age of 82.

Because this result contains Schwarz's Lemma as a special case [we shall clarify this shortly] it is often called the *Schwarz–Pick Lemma*. Despite the startling nature of the result, we can actually understand its essence very simply; we need only ask the question, “How do the *Beltrami–Poincarites* view [7.15]?”

Because their concept of angle is identical to ours, it follows that their concept of an analytic function is also the same as ours— f appears conformal both to us and to them. In addition, we both agree that rays emanating from the origin are straight lines along which we may measure distance. Consequently, the Beltrami–Poincarites willingly concede that w is closer to 0 than z is, although they violently disagree with our quantitative determination of exactly *how much* closer it is. Now recall that there is a small (psychological) flaw in the Beltrami–Poincaré model: 0 appears special to us because it is the centre of the disc, but to the Beltrami–Poincarites who inhabit an infinite, homogeneous plane 0 is *utterly indistinguishable from any other point of their world*.

The above explanation is formalized in [7.16]. In the top left figure we see that the Beltrami–Poincarites have marked an arbitrary point a , drawn a few concentric circles centred there, and on the outermost of these they have marked a second point b . They (and we) now consider the effect of an analytic mapping f of their world to itself. The point a is sent to some image point $A = f(a)$ [top right] and likewise b is sent to $B = f(b)$. In order to compare the separation of A and B with that of a and b , the Beltrami–Poincarites perform a rigid motion ($M_A \circ M_a$ would do) that moves the circles centred at a to circles (of equal hyperbolic size) centred at A . Consequently, the hyperbolic separation of a and b will have been decreased [increased] by f according as B lies inside [outside] the outermost of these circles. In anticipation of Pick's result we have drawn it inside, corresponding to a decrease in hyperbolic separation. However, observe that as in our previous numerical example, to us Euclideans it looks as though the separation has been *increased*.

In order to show us poor blind Euclideans that the circles centred at a and A really are concentric and of equal sizes (so enabling us to see that B really has gotten closer) the Beltrami–Poincarites perform the illustrated rigid motions M_a and M_A . These respectively move a and A to the origin [bottom left and bottom right figures], yielding circles that are as concentric to us as they always were to them. M_a moves b to $z = M_a(b)$, while M_A moves B to

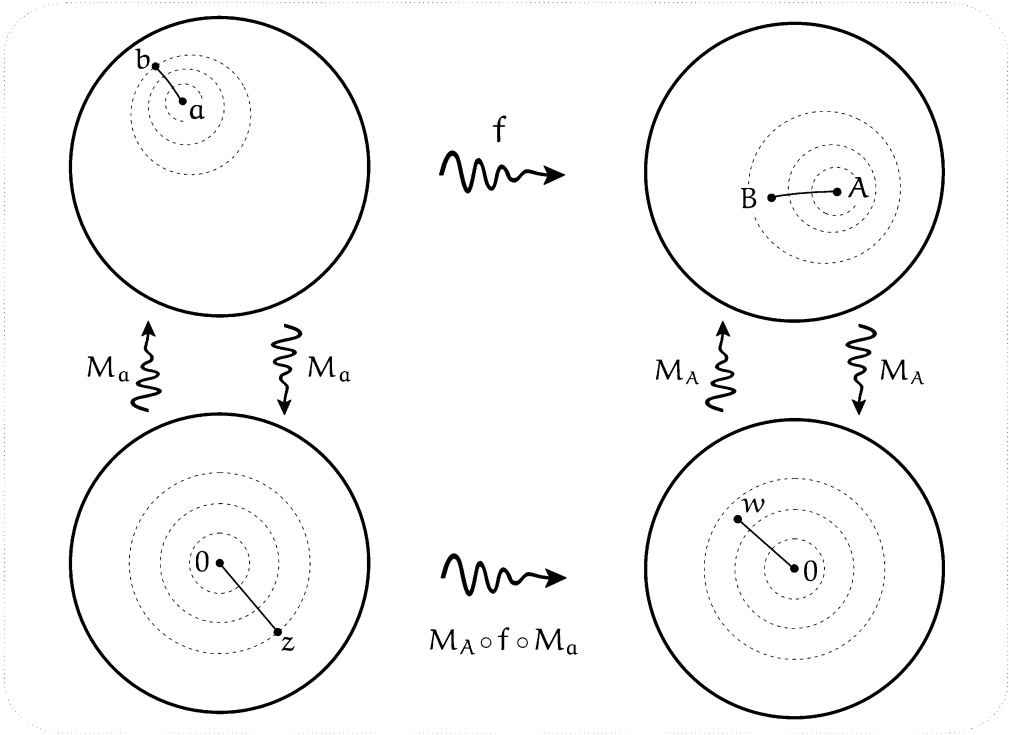
$$w = M_A(B) = (M_A \circ f)(b) = (M_A \circ f \circ M_a)(z).$$

We shall abbreviate $(M_A \circ f \circ M_a)$ to F , so that $w = F(z)$.

We can now see that the following are all equivalent:

$$\mathcal{H}\{A, B\} < \mathcal{H}\{a, b\} \iff \mathcal{H}\{0, w\} < \mathcal{H}\{0, z\} \iff |w| = |F(z)| < |z|.$$

But F is an analytic mapping of the disc to itself which leaves the origin fixed, and so it is subject to Schwarz's Lemma. Thus unless F is a rotation—in which case



[7.16] The Schwarz–Pick Lemma is Pick’s brilliant generalization of Schwarz’s Lemma, stating that the hyperbolic separation of every pair of points must decrease. The Beltrami–Poincarites (our imaginary inhabitants of the Beltrami–Poincaré disc) have no way of distinguishing the bottom half of this figure [aka the already-established Schwarz’s Lemma] from the top half, which is Pick’s result. Q.E.D.!

$f = (M_A \circ F \circ M_a)$ is a rigid motion—we must have $|w| = |F(z)| < |z|$, as depicted. Done.

Finally, let us express the Schwarz–Pick Lemma in symbolic form. If f is not a rigid motion then $|F(z)| < |z|$, which may be written out more explicitly as

$$|(M_A \circ f \circ M_a)(z)| < |z|,$$

which in turn can be written as

$$|(M_A \circ f)(b)| < |M_a(b)|.$$

Thus

$$\left| \frac{B - A}{\overline{A}B - 1} \right| < \left| \frac{b - a}{\overline{a}b - 1} \right|.$$

If we move b closer and closer to a , then $da = (b - a)$ becomes an infinitesimal vector emanating from a whose image under f is an infinitesimal vector $dA = (B - A)$ emanating from A . The above inequality now becomes

$$\frac{|dA|}{1 - |A|^2} < \frac{|da|}{1 - |a|^2},$$

which we may interpret [cf. (6.44), p. 363] as saying that, provided f is not a rigid motion, the hyperbolic length of dA is less than that of its preimage da . This is the infinitesimal version of (7.15).

7.8 The Generalized Argument Principle

7.8.1 Rational Functions

We have now seen that there are many powerful and surprising consequences of the Topological Argument Principle as restricted to analytic functions. Still others are described in the exercises. However, in all our previous work we have only examined mappings which are free of singularities in the region under consideration. We now lift this restriction and find that there is a generalization of (7.6) which applies to this case also.

We began our discussion of the Argument Principle by looking at the prototypical analytic functions without singularities—the polynomials. In order to understand the generalization to analytic functions *with* singularities, we should correspondingly begin with rational functions.

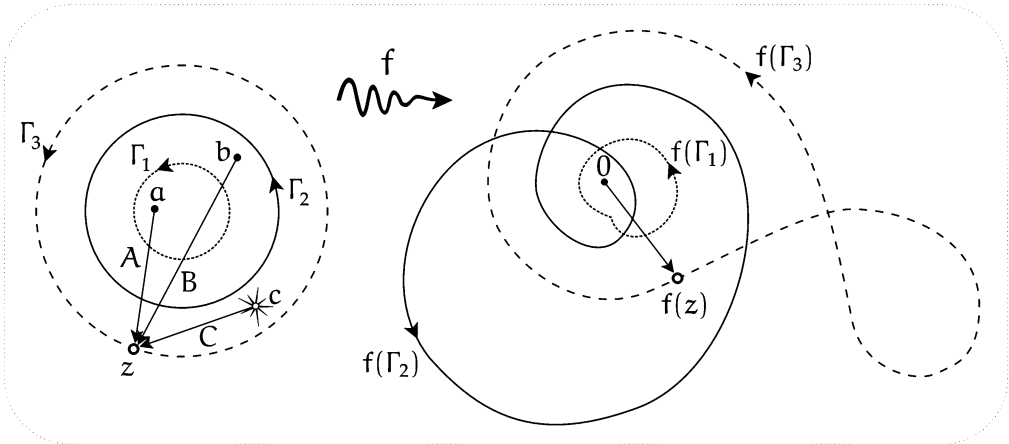
As in [7.8], let A , B , and C be the complex numbers from the fixed points a , b , and c to the variable point z . The left-hand side of [7.17] shows an expanding circle Γ at three successive stages of its growth: Γ_1 , Γ_2 , and Γ_3 . The right-hand side shows the evolution of the image of Γ under the rational mapping

$$f(z) = \frac{(z - a)(z - b)}{(z - c)} = A B \cdot \frac{1}{C}. \quad (7.16)$$

By the time Γ has grown into Γ_1 it has enclosed a , and $v[f(\Gamma_1), 0] = 1$, in accord with the ordinary Argument Principle. As Γ continues to grow it crosses the other 0-point at b , and the winding number of $f(\Gamma)$ correspondingly increases to $v[f(\Gamma_2), 0] = 2$. Now comes the new phenomenon. As Γ crosses the singularity at c the winding number of its image *decreases* by one so that $v[f(\Gamma_3), 0] = 1$.

The explanation is simple. As z traverses Γ_3 , the winding number of $f(z)$ is the sum of the revolutions executed separately by A , B , and $(1/C)$. The first two go round once, but as C rotates counterclockwise, $(1/C)$ rotates in the opposite direction, finally executing one complete *negative* revolution. By the same token, if the denominator of f instead contained C^m then $(1/C^m)$ would execute $-m$ revolutions, and the winding number would become

$$v[f(\Gamma_3), 0] = 2 - m.$$



[7.17] The Generalized Argument Principle. Consider $f(z) = (z - a)(z - b) \cdot \frac{1}{(z - c)} = A B \cdot \frac{1}{C}$. Traversing Γ_1 , A executes a complete revolution, while B and C merely oscillate, so $\nu[f(\Gamma_1), 0] = 1$. Expanding Γ to enclose b, both A and B now execute complete revolutions, while C merely oscillates, so $\nu[f(\Gamma_2), 0] = 2$. But when Γ expands still further to Γ_3 , enclosing the pole at c (marked by the explosion!) then $(1/C)$ executes a negative revolution, decreasing the winding number by one: $\nu[f(\Gamma_3), 0] = 1$. In general, if N and M are the number of interior p-points and poles, both counted with their multiplicities, then $\nu[f(\Gamma), p] = N - M$.

As with counting zeros, we could say in this case that c was a singularity [or *pole*, as we shall now call such places] of multiplicity m.

The previous equation is an example of the Generalized Argument Principle:

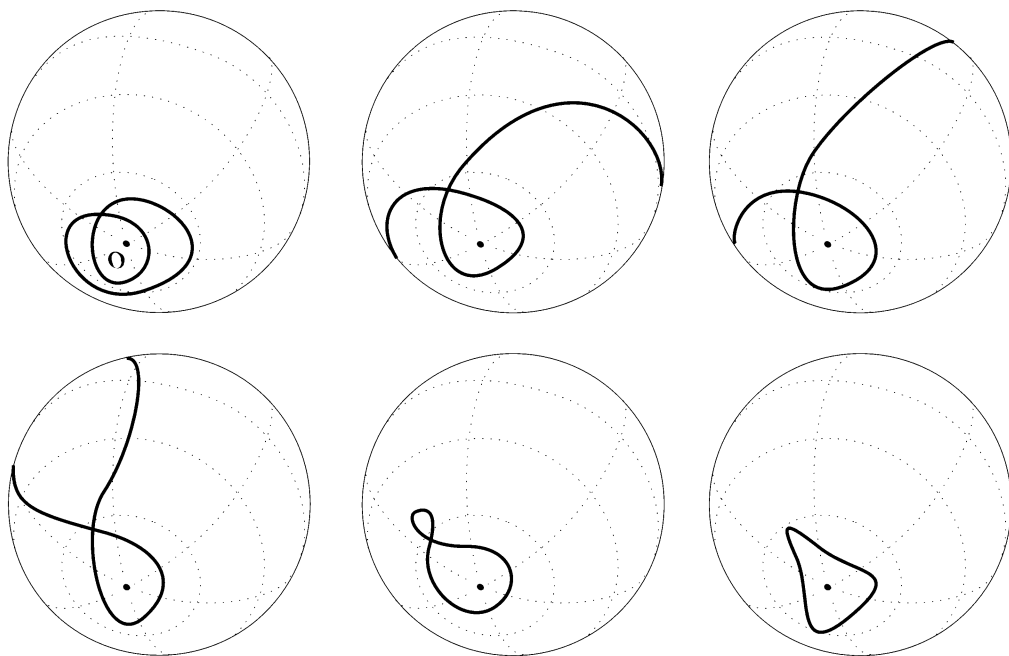
Let f be analytic on a simple loop Γ and analytic inside except for a finite number of poles. If N and M are the number of interior p-points and poles, both counted with their multiplicities, then $\nu[f(\Gamma), p] = N - M$. (7.17)

Simply by allowing an arbitrary number of factors on the top and bottom of (7.16) we see that this result is certainly true when f is any rational function.

Before explaining why it works in general, let us develop a more vivid understanding of how it works in the case of our example (7.16). We have certainly shown that as Γ crosses c the winding number drops from 2 to 1, but exactly *how* does this unwinding occur?

If we look at the image plane just as Γ crosses c then $f(\Gamma)$ undergoes a sudden and violent change of shape as it leaps to infinity and then returns, but this leaves us none the wiser. However, if we instead watch its evolution on the *Riemann sphere* then we gain a new and delightful insight into the process.

Figure [7.18] (which should be scanned like a comic strip) illustrates this. At the time of the first picture [top left] Γ has already enclosed the two roots, and its image is seen to wind round the origin twice. Now follow the evolution of $f(\Gamma)$



[7.18] Reducing the winding number by passing over the point at infinity. *The Riemann sphere provides a vivid explanation of the previous figure. As Γ expands from Γ_2 to Γ_3 , crossing the pole at c , the loop on the Riemann sphere simply slides over the north pole, unwinding the loop and reducing its winding number from 2 to 1.*

through the remaining pictures. As Γ crosses c [top right] there is no longer any excitement— $f(\Gamma)$ merely slides across the north pole, and *this* is how the unwinding is achieved. Try using a computer to animate the evolution of the image $f(\Gamma)$ on the Riemann sphere as Γ expands through the roots and poles of a rational function f of your choosing.

7.8.2 Poles and Essential Singularities

In generalizing the ordinary Argument Principle we had to ask ourselves how we should count p -points of a general analytic function. The factorization (7.7) brought out the analogy with polynomials and gave us a satisfactory definition of the algebraic (and topological) multiplicity of a p -point.

The method of extending (7.17) from rational functions to analytic functions with singularities is essentially the same. The only complication is that there are actually *two* possible kinds of singularity for an otherwise analytic function.

The first kind of singularity is called a *pole*. It is by far the most commonly encountered type in applications of complex analysis, and it is the only type to which (7.17) applies. Here's the definition. If $f(z)$ approaches ∞ as z approaches a from any direction then a is a pole of f . We can understand the terminology by

thinking of the modular surface of f , for there will be an infinitely high spike or “pole” above the point a . Figure [2.14] on p. 74 is an example of this.

Since f is analytic, it follows that $F(z) \equiv [1/f(z)]$ is also analytic and has a root at a . If this root has multiplicity m then the factorization (7.7) of F is

$$F(z) = (z - a)^m \Omega(z), \quad (7.18)$$

where Ω is analytic and nonzero at a ; in fact we know that $\Omega(a) = F^{(m)}(a)/m!$. The local behaviour of f near a is therefore given by

$$f(z) = \frac{\tilde{\Omega}(z)}{(z - a)^m}, \quad (7.19)$$

where $\tilde{\Omega}(z) = [1/\Omega(z)]$ is analytic and nonzero at a . This expression brings out the analogy with rational functions and enables us to identify m as the algebraic multiplicity or *order* of the pole at a . We call a pole *simple*, *double*, *triple*, etc., according as $m = 1, 2, 3$, etc.

Note that we have also found a way of calculating the order of a pole, namely, as the order of the first nonvanishing derivative of $(1/f)$. Once you have identified the locations of the poles, you may use this method [exercise] to find the orders of the poles of the following functions:

$$P(z) = \frac{1}{\sin z}; \quad Q(z) = \frac{\cos z}{z^2}; \quad R(z) = \frac{1}{(e^z - 1)^3}.$$

You should have found that P has a simple pole at each multiple of π ; Q has a double pole at 0; and R has a triple pole at each multiple of $2\pi i$.

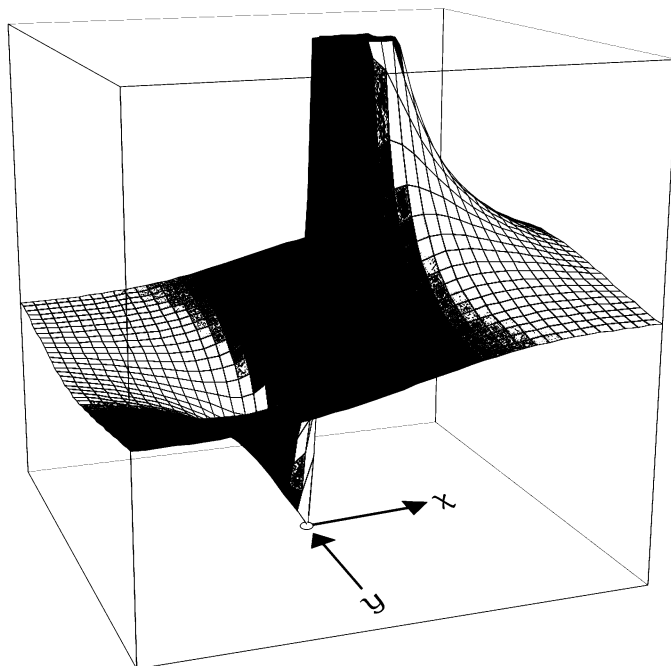
One more piece of terminology. If the only singularities in some region of an otherwise analytic function are poles, the function is called *meromorphic* in that region.

In addition to poles, it is also possible for an otherwise analytic function to possess what are called *essential singularities*. We shall postpone detailed discussion of such places to a later chapter, but it is clear that the behaviour of a function f in the vicinity of an essential singularity s must be very strange and wild. If f were bounded in the vicinity of s then s would not be a singularity at all, but on the other hand $f(z)$ cannot approach ∞ as z approaches s from all directions, for then s would only be a pole.

Consider the standard example $g(z) = e^{1/z}$, which clearly has a singularity of some type at the origin. If we write $z = r e^{i\theta}$ then

$$|g(z)| = e^{\frac{\cos \theta}{r}}.$$

Figure [7.19] depicts the modular surface. If z approaches 0 along the imaginary axis then $|g(z)| = 1$. But if the approach is instead made along a path lying to the left of the imaginary axis (where $\cos \theta < 0$) then $g(z)$ tends to 0. Finally, if the



[7.19] The modular surface of $g(z) = e^{1/z}$ reveals its essential singularity at the origin. Since $|g(z)| = e^{\frac{\cos \theta}{r}}$, the behaviour at 0 depends drastically upon the direction in which we approach it. If $\cos \theta < 0$ as we approach, then the height $|g(z)|$ drops to zero, and the surface hits $\mathbb{C}$. If we approach precisely along the tightrope of the imaginary axis (for which $\cos \theta = 0$) then our height holds precariously steady at 1. But if $\cos \theta > 0$ as we approach, then the height explodes to infinity faster than any polynomial!

approach path lies to the right of the imaginary axis then $g(z)$ tends to ∞ . In fact, not only will $|g(z)|$ become infinite in this case, but the *rate* at which it zooms off to ∞ is quite beyond the ken of any pole.

To see this, reconsider (7.19). The greater the order m , the faster the growth of f as the pole at a is approached. However, no matter how great the order happens to be, we know that $(z - a)^m$ dies away fast enough to kill this growth, in the sense that its product with f remains bounded. Indeed, the order of a pole can be *defined* as the smallest power of $(z - a)$ which will curb the growth of f in this way.

Compare this with the growth of $g(z)$ as its essential singularity is approached, say along the positive real x -axis. To confirm that g grows faster than any meromorphic function, we need only recall from ordinary calculus that

$$\lim_{x \rightarrow 0} x^m e^{1/x} = \lim_{\lambda \rightarrow \infty} \frac{e^\lambda}{\lambda^m} = \infty,$$

no matter how great the value of m .

7.8.3 The Explanation*

In order to explain (7.17) let us return to the interpretation of (7.19). If we think of f as mapping into the Riemann sphere Σ , then the north pole (∞) is an image point like any other, and the poles of f are simply its preimages, ∞ -points if you will. As we now explain, this means that the topological multiplicity of an ∞ -point can be defined in exactly the same way as that of any other p -point, namely, as the number of times that the image of a small loop round a winds round $f(a)$.

Reconsider the mapping F in (7.18). By virtue of (7.8), we know that a sufficiently small circle C_a centred at a will be mapped to a small loop $F(C_a)$ winding round the origin m times. On Σ , the stereographic projection of $F(C_a)$ therefore winds round the south pole m times, counterclockwise as seen from *inside* Σ . Because complex inversion (which sends $F(C_a)$ to $f(C_a) = 1/[F(C_a)]$) rotates Σ about the real axis by π , thereby swapping 0 and ∞ , this means that $f(C_a)$ will be a small loop winding m times round ∞ . Since it winds counterclockwise as seen from inside Σ , its stereographic projection in the plane is therefore a very large loop winding m times *clockwise* around the origin, i.e., with winding number $-m$.

As an aside, observe that it now makes sense to rewrite (7.19) as

$$f(z) = (z - a)^{-m} \tilde{\Omega}(z),$$

and to correspondingly think of a pole of order m as being a *root of negative multiplicity* $-m$.

Shifting our attention away from the origin, we now consider winding numbers around an arbitrary (finite) point p . By making C_a sufficiently small, we can be certain that $f(C_a)$ will be so large that it will wind $-m$ times round p . But if we expand C_a into any simple loop Γ_a without crossing any p -points or other poles, then the winding number of the image round p cannot change. In other words,

$$\text{If } a \text{ is a pole of order } m \text{ and } \Gamma_a \text{ is any simple loop containing } a \text{ but no } p\text{-points and no other poles, then } \nu[f(\Gamma_a), p] = -m. \quad (7.20)$$

Finally, reconsider figure [7.11]. You may now easily convince yourself that the argument leading to (7.11) remains valid if some of the a_j 's are poles instead of p -points. Let's call these singular points s_j . Thus

$$\nu[f(\Gamma), p] = \sum_{p\text{-points}} \nu[f(\Gamma_{a_j}), p] + \sum_{\text{poles}} \nu[f(\Gamma_{s_j}), p].$$

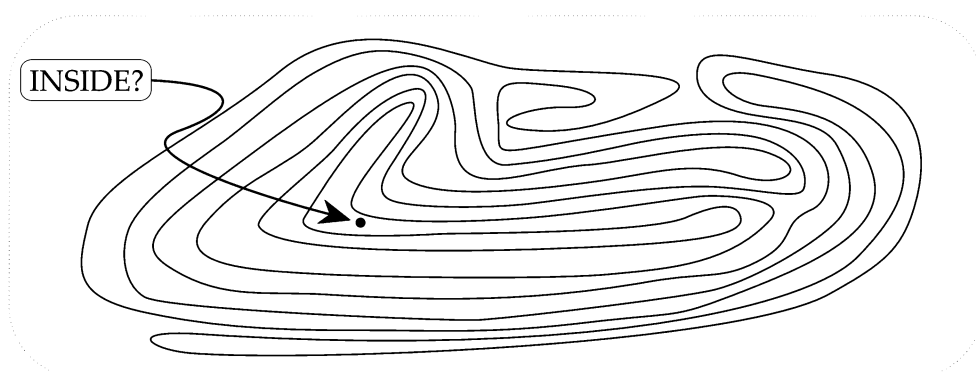
Using (7.20), this implies that

$$\nu[f(\Gamma), p] = [\text{number of } p\text{-points inside } \Gamma] - [\text{number of poles inside } \Gamma],$$

as was to be shown.

7.9 Exercises

- 1 A “simple” loop can get very complicated (see diagram). However, if we imagine creating this complicated loop by gradually deforming a circle, it is clear that it will wind round its interior points precisely once. Let $N(p)$ be the number of intersection points of the simple loop with a ray emanating from p (cf. [7.4]). What distinguishes the possible values of $N(\text{interior point})$ from those of $N(\text{exterior point})$? In place of the crossing rule (7.1), you now possess (for simple loops) a much more rapid method of determining whether a point is inside or outside.



You can use this result to play a trick on a friend F : (1) So that foul play cannot be suspected, get F to draw a very convoluted simple loop for himself; (2) choose a random point in the thick of things and ask F if it's inside or not, i.e., starting at this point, can one escape through the maze to the outside?; (3) after F has been forced to recognize the time and effort required to answer the question, get him to choose a point for you; (4) choosing a ray in your mind's eye, scan along it and count the intersection points. Amaze F with your virtually instantaneous answer!

- 2 Reconsider the mapping $\hat{\mathcal{L}}$ in (7.4) of the unit circle to itself, and the associated graph of $\Phi(\theta)$ in [7.7]. If $\Phi'(a) > 0$ then the graph is rising above the point $\theta = a$, and small movement of z will produce a small movement of the w having the *same sense*. We say that Φ is orientation-preserving at a and that the topological multiplicity $\nu(a)$ of $z = e^{ia}$ as a preimage of $w = e^{i\Phi(a)}$ is $+1$. Similarly, if $\Phi'(a) < 0$ then the mapping is orientation-reversing and $\nu(a) = -1$. In other words,

$$\nu(a) = \text{the sign of } \Phi'(a).$$

Compare this with the 2-dimensional formula (7.9).

- (i) In [7.7], explain how the complete set of preimages of $w = e^{iA}$ can be found by drawing the family of horizontal lines $\Phi = A, A \pm 2\pi, A \pm 4\pi$, etc.

- (ii) If the set of preimages is typical in the sense that $\Phi' \neq 0$ at any of them, what do we obtain if we sum their topological multiplicities? Thus to say that the degree of $\widehat{\mathcal{L}}$ (the winding number of L) is ν is essentially to say that $\widehat{\mathcal{L}}$ is ν -to-one. [Hint: In [7.4], consider a ray as describing the location of w .]
- 3 For each of the following functions $f(z)$, find all the p -points lying inside the specified disc, determine their multiplicities, and by using a computer to draw the image of the boundary circle, verify the Argument Principle.
- (i) $f(z) = e^{3\pi z}$ and $p = i$, for the disc $|z| \leq (4/3)$.
 - (ii) $f(z) = \cos z$ and $p = 1$, for the disc $|z| \leq 5$.
 - (iii) $f(z) = \sin z^4$ and $p = 0$, for the disc $|z| \leq 2$.
- 4 Reconsider [7.8].
- (i) Use a computer to draw the image under a cubic mapping

$$f(z) = (z - a)(z - b)(z - c)$$

of an expanding circle Γ , and observe the *manner* in which the winding number increases as Γ passes through the roots a , b , and c . In particular, observe that the shape that marks the birth of a new loop is this: $\prec$.

- (ii) If $f'(p) \neq 0$ then a little piece of Γ passing through p is merely amplitwisted to another almost straight piece of curve through $f(p)$. Deduce that $\prec$ shapes can only occur when Γ hits a critical point. Explain why the particular shape $\prec$ is consistent with a critical point of order 1.
- (iii) Observe that there are only two point in the evolution of Γ at which a $\prec$ shape is produced. Explain this algebraically in terms of the degree of f' .
- (iv) Let T be the triangle with vertices a , b , and c . There are many ellipses which can be inscribed in T so as to touch all three sides, but show that there is only one (call it $\mathcal{E}$) that touches T at the *midpoints* of the sides.

Hard Show that the two critical points of f are the *foci* of $\mathcal{E}$!

- 5 As in the text, let ξ_a , η_a , ϕ_a denote the two perpendicular expansion factors and the rotation angle used to describe the SVD decomposition of the local linear transformation at a produced by a mapping. By considering the case of a rotation by $(\pi/4)$, for which J is constant, show that ξ and η are generally *not* the eigenvalues λ_1 and λ_2 of the Jacobian J . However, confirm for this example that $\det(J) = \lambda_1\lambda_2 = \xi\eta$.
- 6 Even in three or more dimensions the local linear transformation induced by a mapping f at a point a can still be represented by the Jacobian matrix $J(a)$, and if a is not a critical point then its topological multiplicity $\nu(a)$ as a preimage

of $f(a)$ is still given by (7.9). If n is the number of real negative eigenvalues of $J(a)$, counted with their algebraic multiplicities, show that

$$\nu(a) = (-1)^n.$$

[Hint: Since the characteristic equation $\det[J(a) - \lambda I] = 0$ has real coefficients, any complex eigenvalues must occur in conjugate pairs.]

- 7 Consider the nonanalytic mapping $h(z) = |z|^2 - i\bar{z}$.
- (i) Find the roots of h .
 - (ii) Calculate the Jacobian J , and hence find $\det(J)$.
 - (iii) Use (7.9) to calculate the multiplicities of the roots in (i).
 - (iv) Find the image curve traced by $h(z)$ as $z = 2e^{i\theta}$ traverses the circle $|z| = 2$, and confirm the prediction of the Topological Argument Principle.
 - (v) Gain a better understanding of the above facts by observing that $h(z) = \bar{z}(z - i)$, and then mimicking the analysis of [7.8].
 - (vi) Use the insight of the previous part to find $\nu(i/2)$, which cannot be done with (7.9).
- 8 Let $Q(t)$ be a real function of time t , subject to the differential equation

$$c_n \frac{d^n Q}{dt^n} + c_{n-1} \frac{d^{n-1} Q}{dt^{n-1}} + \cdots + c_1 \frac{dQ}{dt} + c_0 Q = 0.$$

Recall that one solves this equation by taking a linear superposition of special solutions of the form $Q_j(t) = e^{s_j t}$. Substitution into the previous equation shows that the s_j 's are the roots of the polynomial

$$F(s) \equiv c_n s^n + c_{n-1} s^{n-1} + \cdots + c_1 s + c_0.$$

Note that $Q_j(t)$ will decay with time if s_j has a negative real part. The issue of whether or not the general solution of the differential equation decays away with time therefore reduces to the problem of determining whether or not all n roots of $F(s)$ lie in the half-plane $\operatorname{Re}(s) < 0$. Let $\mathcal{R}$ be the net rotation of $F(s)$ as s traverses the imaginary axis from bottom to top. Explain the following result: *The general solution of the differential equation will die away if and only if*

$$\mathcal{R} = n\pi.$$

This is called the *Nyquist Stability Criterion*, and an F that satisfies this condition is called a *Hurwitz polynomial*. [Hints: Apply the Argument Principle to the loop consisting of the segment of the imaginary axis from $-iR$ to $+iR$, followed by one of the two semicircles having this segment as diameter. Now let R tend to infinity.]

9 Referring to the previous exercise, consider the differential equation

$$\frac{d^3 Q}{dt^3} - Q = 0.$$

- (i) Find $\mathcal{R}$ for this equation. Does it satisfy the Nyquist Stability Criterion?
- (ii) Confirm your conclusion by explicitly solving the differential equation.

10 If a is real and greater than 1, use Rouché's Theorem to show that the equation

$$z^n e^a = e^z$$

has n solutions inside the unit circle.

11 (i) Applying Rouché's Theorem to $f(z) = 2z^5$ and $g(z) = 8z - 1$, show that all five solutions of the equation $2z^5 + 8z - 1 = 0$ lie in the disc $|z| < 2$.

(ii) By reversing the roles of f and g , show that there is only one root in the unit disc. Deduce that there are four roots in the ring $1 < |z| < 2$.

12 We can formalize, and slightly generalize, our explanation of Rouché's Theorem as follows:

- (i) If $p(z)$ and $q(z)$ are nonzero on a simple curve Γ , and $\tilde{\Gamma}$ is the image curve under $z \mapsto p(z)q(z)$, show that

$$\nu[\tilde{\Gamma}, 0] = \nu[p(\Gamma), 0] + \nu[q(\Gamma), 0].$$

- (ii) Write

$$f(z) + g(z) = f(z) \left[1 + \frac{g(z)}{f(z)} \right] = f(z) H(z).$$

If $|g(z)| < |f(z)|$ on Γ , sketch a typical $H(\Gamma)$. Deduce that

$$\nu[H(\Gamma), 0] = 0.$$

Using the previous part, obtain Rouché's Theorem.

- (iii) If we only stipulate that $|g(z)| \leq |f(z)|$ on Γ , then parts of $H(\Gamma)$ could actually coincide with the circle $|z - 1| = 1$, rather than lying strictly inside it, and $\nu[H(\Gamma), 0]$ might not be well-defined. However, show that if we further stipulate $f + g \neq 0$ on Γ , then $\nu[H(\Gamma), 0] = 0$ as before. Deduce that $\nu[(f + g)(\Gamma), 0] = \nu[f(\Gamma), 0]$.

13 Let $w = f(z)$ be analytic inside and on a simple loop Γ , and suppose that $f(\Gamma)$ is an origin-centred circle.

- (i) If Δ is an infinitesimal movement of z along Γ and ϕ is the correspondingly infinitesimal rotation of w , show geometrically that

$$\frac{f' \Delta}{f} = i\phi.$$

- (ii) As z traverses Γ , explain why $v[\Delta, 0] = 1$ and $v[i\phi, 0] = 0$.
- (iii) Referring to (i) of the previous exercise, show that

$$v[f(\Gamma), 0] = v[f'(\Gamma), 0] + 1.$$

- (iv) Deduce from the Argument Principle that f has one more root inside Γ than f' has. This is sometimes called *Macdonald's Theorem*, though I believe its essence goes back as far as Riemann.
 - (v) From this we deduce, in particular, that f has at least one root inside Γ . Derive this fact directly by considering the portion of the modular surface lying above Γ and its interior.
- 14 In contrast to analytic mappings, it is perfectly possible for a continuous non-analytic mapping to completely crush pieces of curve or even areas without crushing the rest of its domain. Let us give a concrete example to show that the Topological Argument Principle does not apply to this case. With $r = |z|$, the mapping $h(z) = \phi(r)z$ will be a continuous function of z if $\phi(r)$ is a continuous function of r . Consider the continuous mapping h of the unit disc to itself corresponding to

$$\begin{aligned}\phi(r) &= 0, & 0 \leq r < (1/2); \\ \phi(r) &= 2r - 1, & (1/2) \leq r \leq 1.\end{aligned}$$

- (i) Describe this mapping in visually vivid terms.
 - (ii) Taking Γ to be the circle $|z| = (3/4)$ and letting $p = 0$, try (and fail) to make sense of (7.11).
- 15 The version of Brouwer's Fixed Point Theorem established in the text fell short of the full result in two ways: (A) we assumed that $|g| < 1$ on D rather than $|g| \leq 1$; (B) we essentially used the Topological Argument Principle, which the previous exercise shows to be useless in the general case of a continuous mapping having infinitely many p -points in a finite region. Let's remove these blemishes. Once again let $m(z) = g(z) - z$ be the movement of z , and suppose that Brouwer's result is *false*, so that $m \neq 0$ throughout the disc $|z| \leq 1$. Obtain the desired contradiction as follows:
- (i) By assumption, $m(z) = g(z) - z = g(z) + f(z)$ does not vanish on the unit circle C . Use Ex. 12(iii) to show that if $|g| \leq 1$ then $v[m(C), 0] = 1$.
 - (ii) Let C_r be the circle $|z| = r$, so that $C_1 = C$. By considering the evolution of $v[m(C_r), 0]$ as r increases from 0 to 1, obtain a contradiction with (i).
- The key fact $v[m(C), 0] = 1$ can be obtained more intuitively. Draw a typical movement vector $m(z)$ emanating from z and note that it makes an *acute angle* with the inward unit normal vector $(-z)$ to C , also drawn emanating from z .

But clearly this normal vector undergoes one positive revolution as z traces C . Deduce that the vector m is also dragged round one revolution.

- 16 Let $f(z)$ be an *odd* power of z , and consider its effect on the unit circle C . Note two facts: (1) if p is on C then $f(-p)$ points in the opposite direction to $f(p)$; (2) $\nu[f(C), 0] = \text{odd}$, in particular it cannot vanish. This is only one example of a general result. Show that (1) always implies (2), even if f is merely continuous. [Hints: If f is subject to (1), what can we deduce about the net rotation $\mathcal{R}$ of $f(z)$ as z traverses the semicircle from p to $-p$? How is the rotation produced by the remaining semicircle related to $\mathcal{R}$?]
- 17 Consider a spherical balloon S resting on a plane. If we gradually deflate S , each point will end up on the plane so that we have a continuous mapping H of S into the plane. Observe that the north and south poles, which are antipodal, have the same image. The *Borsuk-Ulam Theorem* says that *any* continuous mapping H of S into the plane will map some pair of antipodal points to the same image. Consider the mapping $F(p) = H(p) - H(p^*)$, where p^* is antipodal to p . The theorem then amounts to showing that F has a root somewhere on S . Prove this. [Hints: It is sufficient to examine the effect of F on just the northern hemisphere. By taking the boundary of this hemisphere (the equator) to be the circle C of the previous exercise, deduce that $\nu[H(C), 0] \neq 0$.]
- 18 Let f be analytic on a simple loop Γ , and let p be a preimage of a point on $f(\Gamma)$ at which $|f|$ is maximum. If ξ is a tangent complex number to Γ at p , and in the same counterclockwise sense as Γ , show geometrically that $\xi f'(p)$ points in the same direction as $if(p)$. What is the analogous result at a positive minimum of $|f|$?
- 19 (i) If p is not a critical point of an analytic function f , show geometrically that the modulus of f increases most rapidly in the direction $[f(p)/f'(p)]$.
- (ii) In terms of the modular surface above p , this direction lies directly beneath the tangent line to the surface having the greatest “slope” (i.e., tan of the angle it makes with the complex plane). Show that the slope of this steepest tangent plane is $|f'(p)|$, and note the analogy with the slope of the ordinary graph of a real function.
- (iii) What does the modular surface look like at a root of order n ?
- (iv) What does the modular surface look like above a critical point of order m ? Using the case $m = 1$, explain why such places are called *saddle points*.
- (v) Rephrase Macdonald’s Theorem [Ex. 13] in terms of the number P of pits and the number S of saddle points in the portion $\mathcal{A}$ of the modular surface lying above Γ and its interior.

- (vi) It is a beautiful fact that, expressed in this form, Macdonald's Theorem can be explained almost purely topologically. The following explanation is adapted from Pólya (1954), though I believe the basic idea goes back to Maxwell and Cayley. Since $|f|$ is constant on Γ , the boundary of $\mathcal{A}$ is a horizontal curve K , and since f is analytic, K is higher than the rest of $\mathcal{A}$. Also, recall that there are no peaks. Suppose for simplicity that f and f' have only simple roots (P and S in number) so that the pits are cone-like, and the saddle points really look like saddles or (more geographically) like mountain passes.

Now imagine a persistent rain falling on the surface $\mathcal{A}$. The pits gradually fill with water and so become P lakes, the depths of which we shall imagine are always equal to each other. What happens to the number of lakes as the water successively rises past each of the S passes? How many lakes are left by the time that the water has finally risen to the level of K ? As required, deduce that

$$P = S + 1.$$

- (vii) Generalize the above argument to roots and critical points that are not simple.
- 20 Let $f(z)$ and $g(z)$ be analytic inside and on a simple loop Γ . By applying the Maximum Modulus Theorem to $(f - g)$, show that if $f = g$ on Γ then $f = g$ throughout the interior.
- 21 Let $\mathcal{R}(L)$ be the net rotation of $f(z)$ round p as z traverses a loop L . For example, if L does not contain p then

$$\nu[L, p] = \frac{1}{2\pi} \mathcal{R}(L).$$

By taking this formula to be the definition of ν , make sense of the statement that the Generalized Argument Principle (7.17) remains valid even if there are some poles and p -points on Γ , provided that we count these points with *half* their multiplicities.

- 22 In [7.11] we used the idea of deformation to derive the argument principle. The figure below shows another method. The interior of Γ , containing various p -points and poles, has been crudely partitioned into cells C_j in such a way that each one contains no more than a single p -point or a single pole. Now think of each cell as being a loop traversed in the conventional direction; this sense is indicated for two adjacent cells in the figure.
- (i) What is the value of $\nu[f(C_j), p]$ if C_j (1) is empty; (2) contains a p -point of order m ; (3) contains a pole of order n ?

(ii) Obtain the Argument Principle by showing that

$$\sum_j \nu[f(C_j), p] = \nu[f(\Gamma), p].$$

[Hint: If an edge of a cell does not form part of Γ then it is also an edge of an adjacent cell, *but traversed in the opposite direction*. What is the net rotation of $f(z)$ round p as z traverses this edge once in one direction, then in the opposite direction?]

